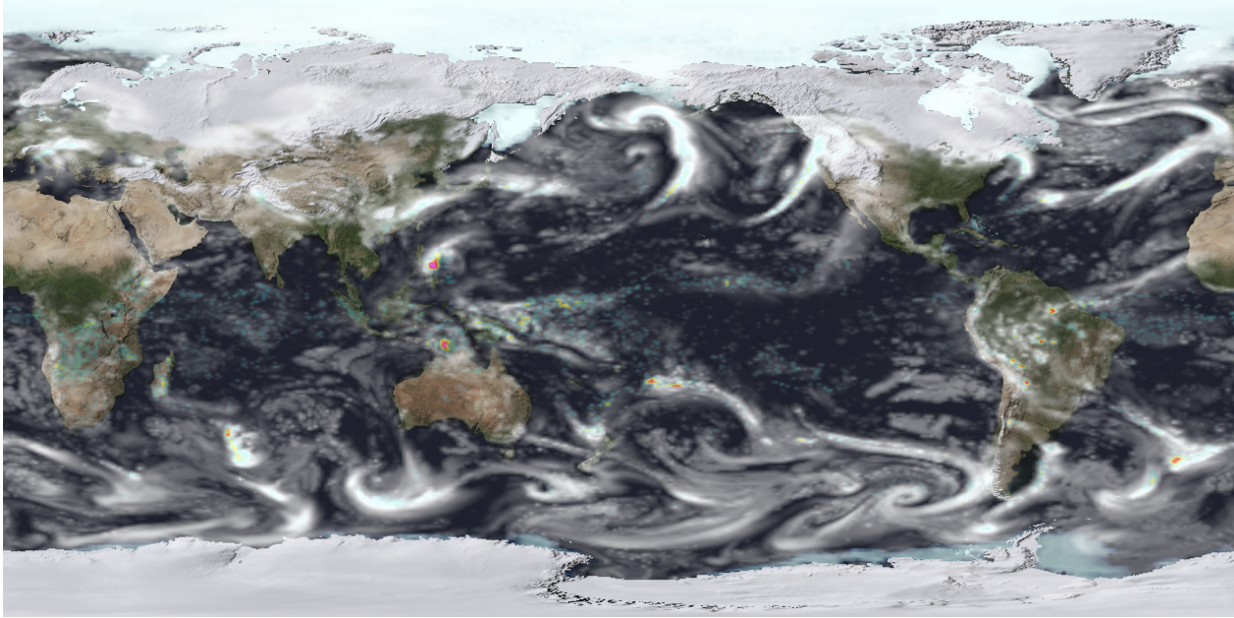


NCAS Climate Modelling Summer School 2019



Elements of dynamics and physics: the ocean

Rémi Tailleux

University of Reading, Reading, UK

Outline

Horizontal (wind-driven?) circulation

- **Some observations**
- **How does momentum from the wind enter the surface of the ocean?**
- **How do the surface layers affect the interior flow**
- **Western boundary currents**

Vertical (density driven?) circulation

- **Some observations**
- **Digression: the equation of state**
- **The overturning circulation**
- **Deep western boundary currents**
- **The importance of mixing**

Outline

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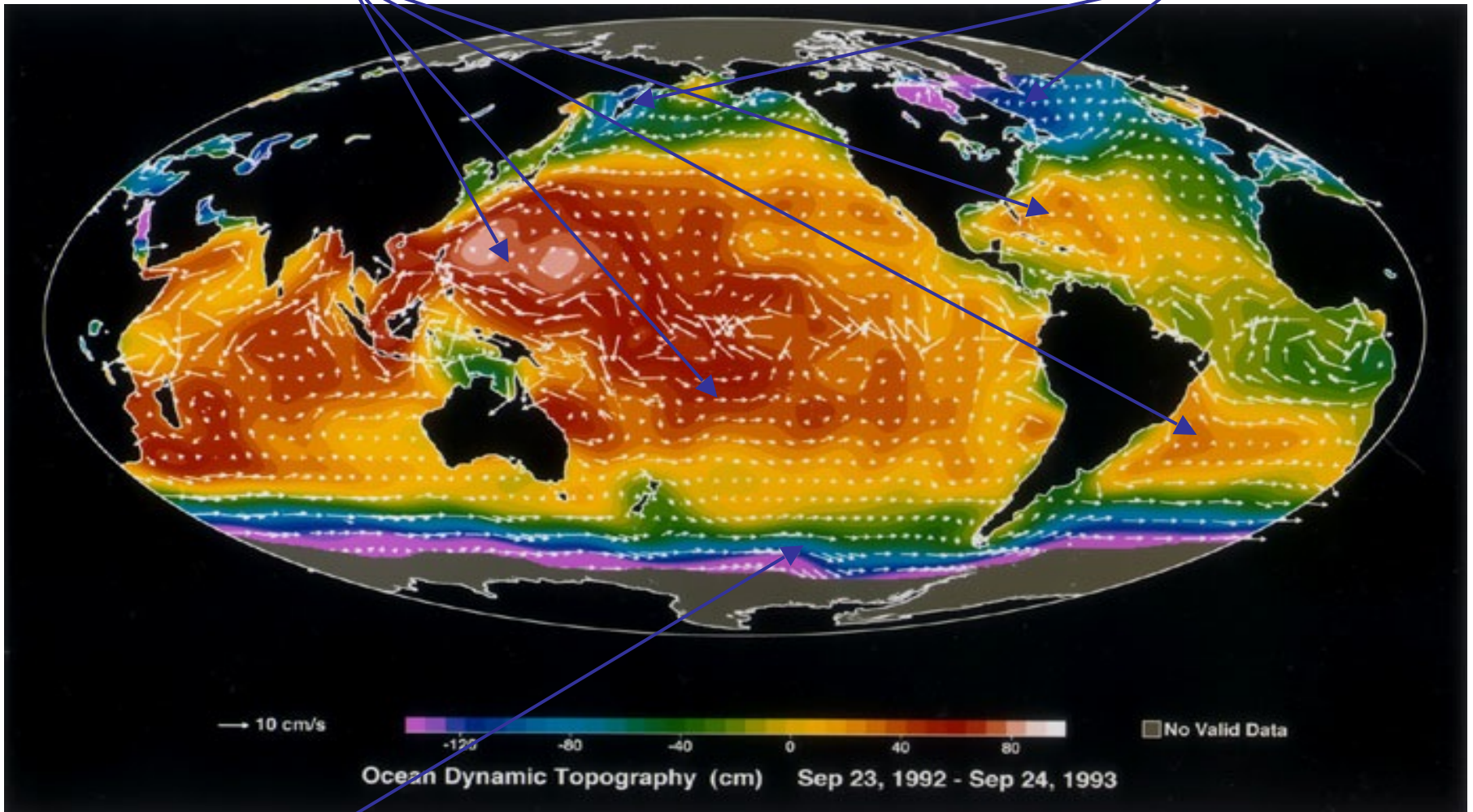
- **Some observations**

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Sea Surface Height Sept 92 – Sept 93

Subtropical max: anticyclonic

Subpolar min: cyclonic

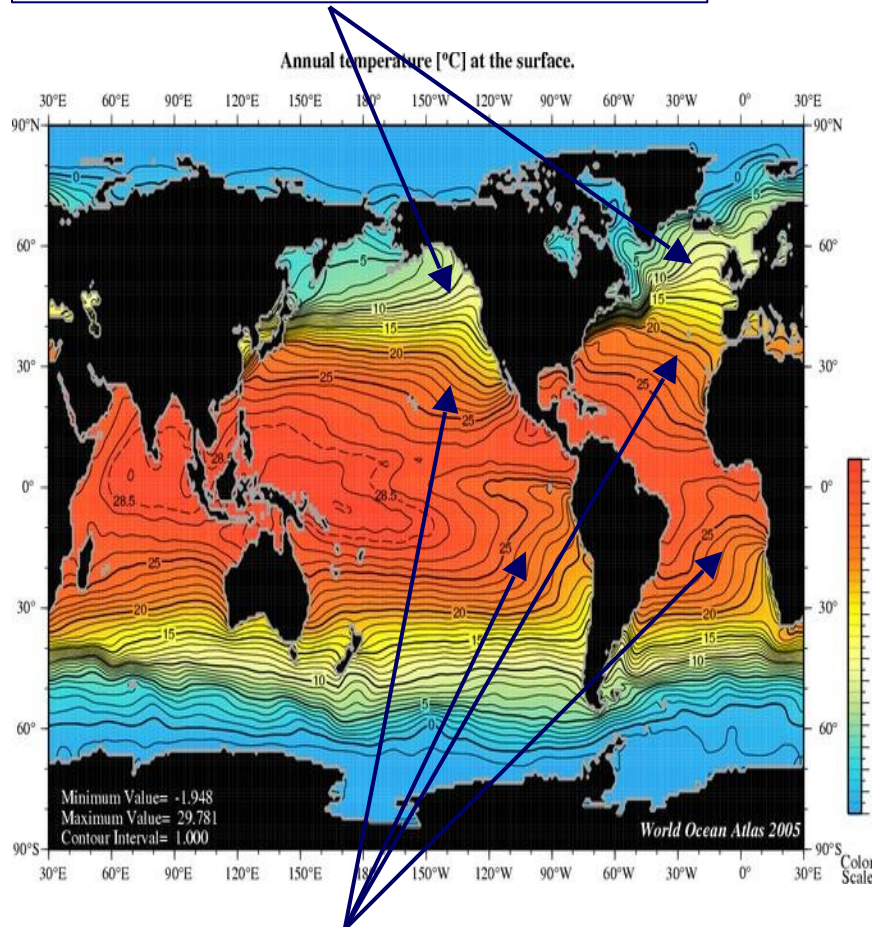


Eastward circumpolar current

From TOPEX/POSEIDON Altimeter, courtesy JPL

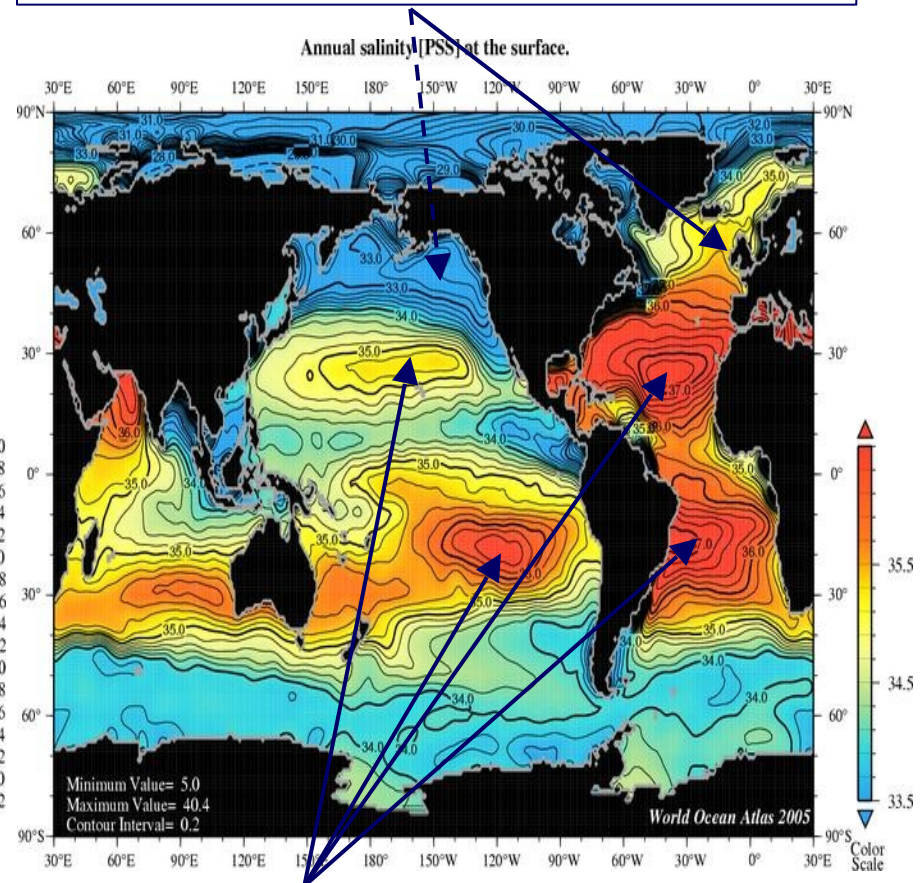
Observed surface temperature and salinity

Isotherms slope poleward to east in subpolar basins



Isotherms slope equatorward to east in subtropics

High salinity 'escapes' northward in N Atlantic, but not N. Pacific



Confined high salinity regions in subtropics. $S_{\text{Atl}} > S_{\text{Pac}}$

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How does momentum from the wind get into the surface layers of the ocean? (The Ekman layer)

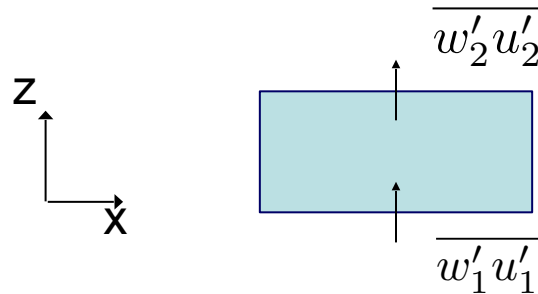
Consider linear momentum equations

$$\begin{aligned}\frac{\partial u}{\partial t} - fv &= -\frac{1}{\rho} \frac{\partial p}{\partial x} + \frac{1}{\rho} \frac{\partial \tau^x}{\partial z} \\ \frac{\partial v}{\partial t} + fu &= -\frac{1}{\rho} \frac{\partial p}{\partial y} + \frac{1}{\rho} \frac{\partial \tau^y}{\partial z}\end{aligned}$$

ρ = density

$\tau_x = -\overline{w'u'}$ = turbulent stress or turbulent momentum transport

$\tau_y = -\overline{w'v'}$ = turbulent stress or turbulent momentum transport



(i.e. parametrise the turbulent motions in terms of the large scale flow)

Introduce closure on momentum mixing

$$\frac{\tau^x}{\rho} = -\overline{w'u'} = K_m \frac{\partial u}{\partial z}$$

and decompose the current into the geostrophic part and a deviation $u=u_g+u_e$. The momentum equations become

$$\begin{aligned} \frac{\partial u_e}{\partial t} - f v_e &= \frac{\partial}{\partial z} \left(K_m \frac{\partial u_e}{\partial z} + K_m \frac{\partial u_g}{\partial z} \right) \\ \frac{\partial v_e}{\partial t} + f u_e &= \frac{\partial}{\partial z} \left(K_m \frac{\partial v_e}{\partial z} + K_m \frac{\partial v_g}{\partial z} \right) \end{aligned}$$

From the thermal 'wind' equation the vertical shear of the geostrophic current is proportional to horizontal density gradient. For near surface large scale flows the geostrophic shear $\ll$ shear of u_e

and $\partial u_g / \partial t \ll \partial u_e / \partial t$ (assumptions).

So

$$\frac{\partial u_e}{\partial t} - f v_e = \frac{\partial}{\partial z} \left(K_m \frac{\partial u_e}{\partial z} \right) \quad (1)$$

$$\frac{\partial v_e}{\partial t} + f u_e = \frac{\partial}{\partial z} \left(K_m \frac{\partial v_e}{\partial z} \right) \quad (2)$$

Take (1)+i (2)

$$\frac{\partial(u_e + i v_e)}{\partial t} + i f(u_e + i v_e) = \frac{\partial}{\partial z} \left(K_m \frac{\partial(u_e + i v_e)}{\partial z} \right)$$

For $\Psi_e = u_e + iv_e \rightarrow$

$$\frac{\partial \Psi_e}{\partial t} + if\Psi_e = \frac{\partial}{\partial z} \left(K_m \frac{\partial \Psi_e}{\partial z} \right)$$

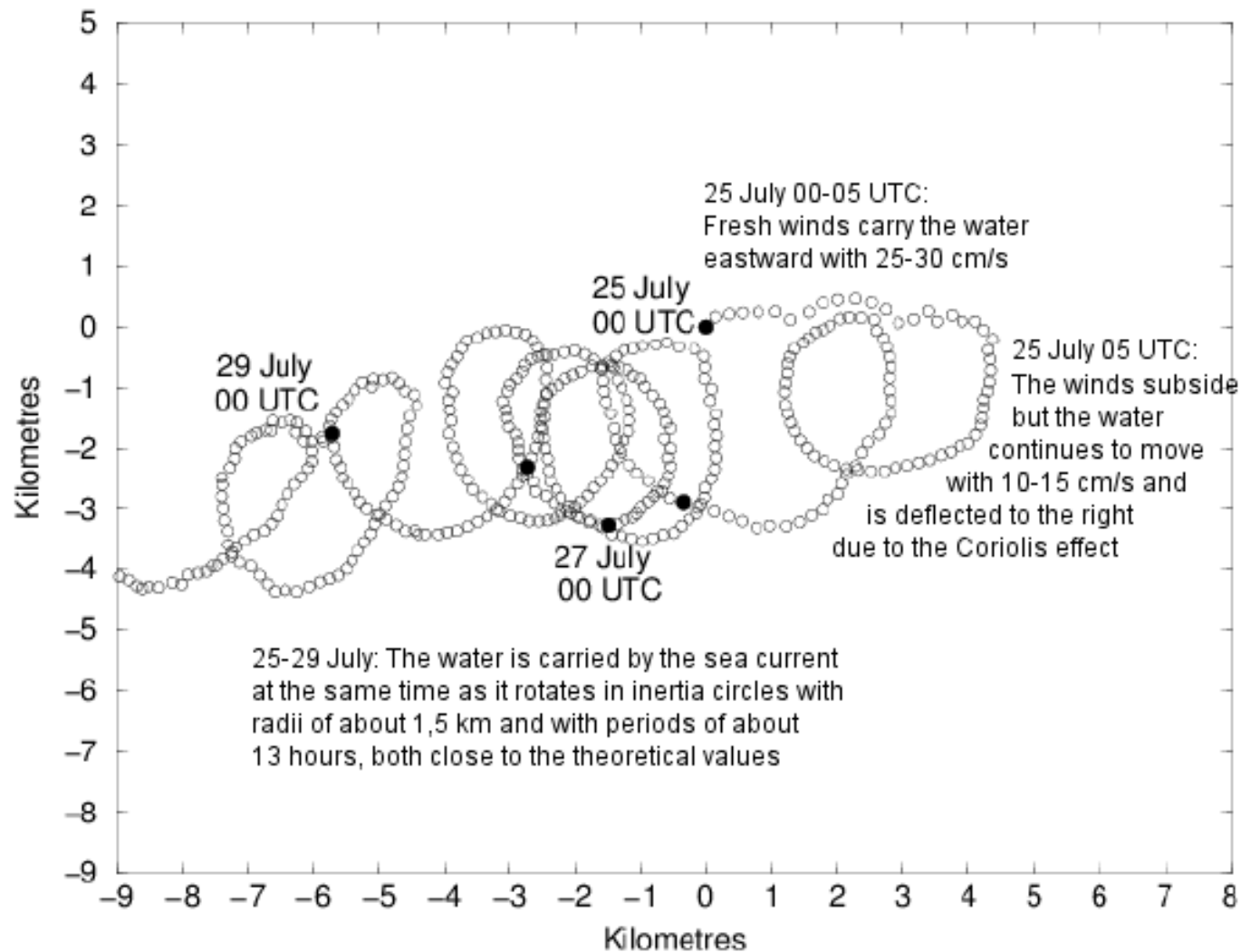
Time-dependent solutions: decaying inertial oscillations

$$\Psi_e \sim e^{-(if+\alpha)t}$$

Steady-state solutions: Ekman spiral of decreasing radius
(solution increasing with depth is unphysical and hence rejected)

$$\Psi_e \sim e^{\sqrt{\frac{f}{2K_m}}z} e^{i\sqrt{\frac{f}{2K_m}}z}$$

Inertial oscillations with period f advected by currents



source: www.cleonis.nl

Properties of Ekman Spiral

Surface velocity: makes angle of 45 deg to the wind (exercise)

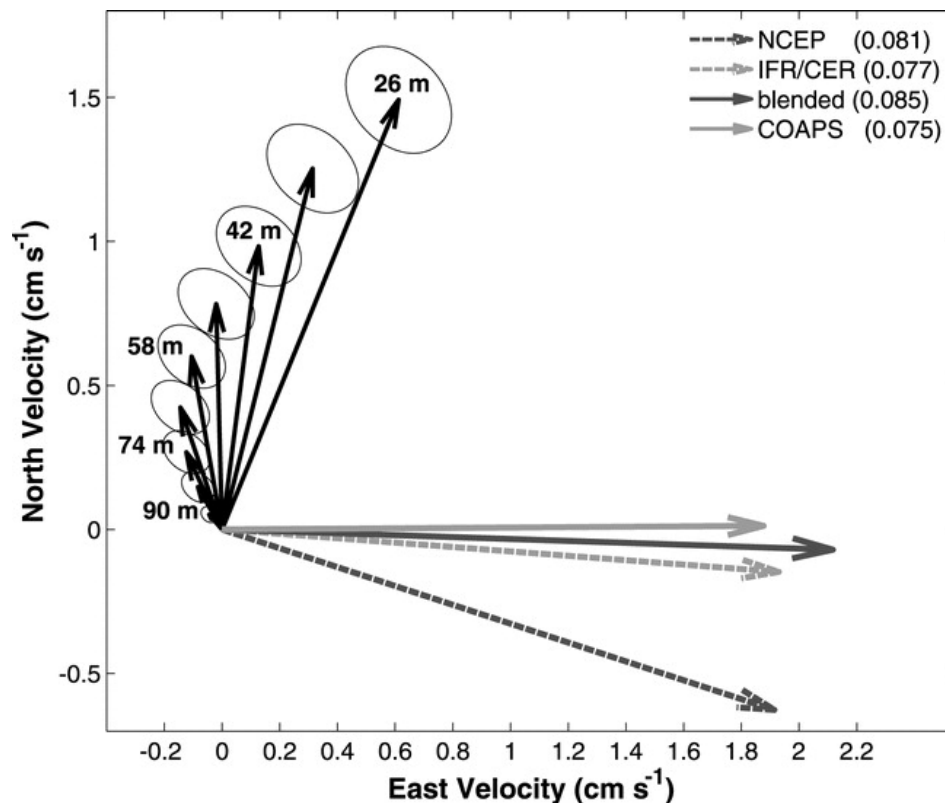
E-folding scale for decaying spiral: Ekman depth

$$h_e = \sqrt{\frac{2K_m}{f}} \quad f = 10^{-4} \text{s}^{-1}, \quad K_m = 0.1 \text{m}^2.\text{s}^{-2} \\ \rightarrow h_e \approx 30 \text{ m}$$

Vertical structure depends sensitively on poorly constrained turbulence properties expressed by K_m

Observations of Ekman spiral: very rare owing to difficulties of separating it from geostrophic shear

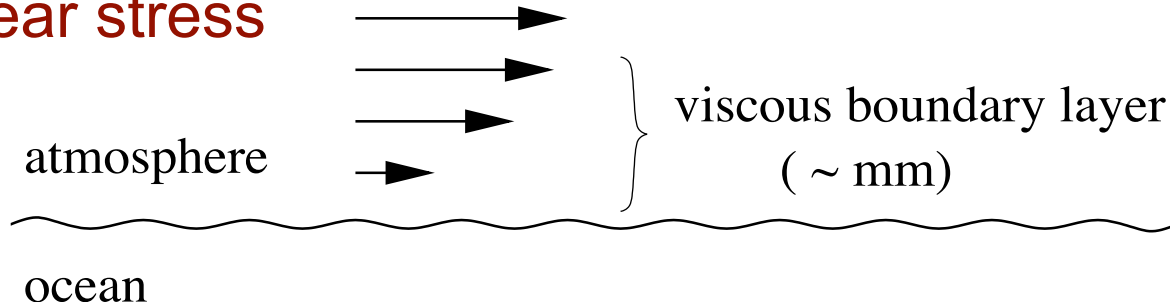
Lenn and Chereskin, 2009 : Observations of Ekman currents in the Southern Ocean, *J. Phys. Oceanogr.*, **39**, 768-779



Quiver plot of observed mean Ekman currents calculated from mean ADCP velocities relative to 98 m. Alternate depths of the Ekman currents are marked and the NCEP (dashed dark gray line), IFR/CER (dashed light gray line), blended (solid dark gray line), and COAPS (solid light gray line) mean wind stresses are shown; the magnitudes of the wind stresses are listed in units of N m^{-2} in parenthesis in the legend. Standard error ellipses are plotted at the tip of each current vector.

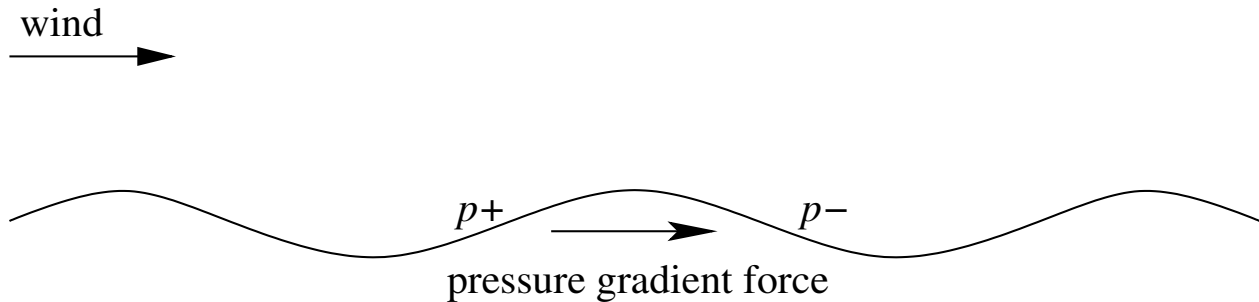
WIND STRESS τ MADE UP OF 2 CONTRIBUTIONS

Viscous shear stress



given by Newton's law of viscosity

Form stress



wind pushes against surface waves $\Rightarrow$ acceleration of fluid parcels

Bulk Formula for Wind Stress

$$\tau_s = C_D \rho_a |\mathbf{U}_{10}| \mathbf{U}_{10}$$

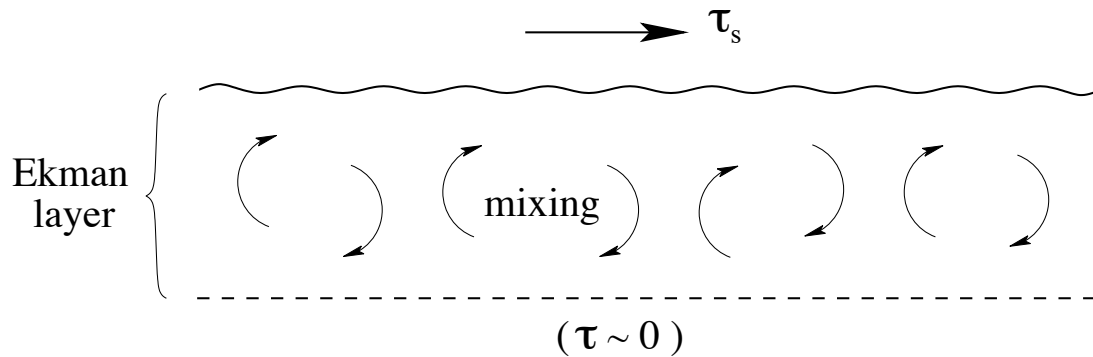
Drag Coefficient
Dimensionless
 $\sim 1.45 \times 10^{-3}$

Air Density
 $\sim 1.2 \text{ kg/m}^3$

10 meters
wind speed
(above sea surface)

Depends somewhat on U10 and surface wave field

Ekman Transport



A key property independent of turbulence details is obtained by integrating steady momentum equations between surface and somewhere below bottom of Ekman layer where stress is small

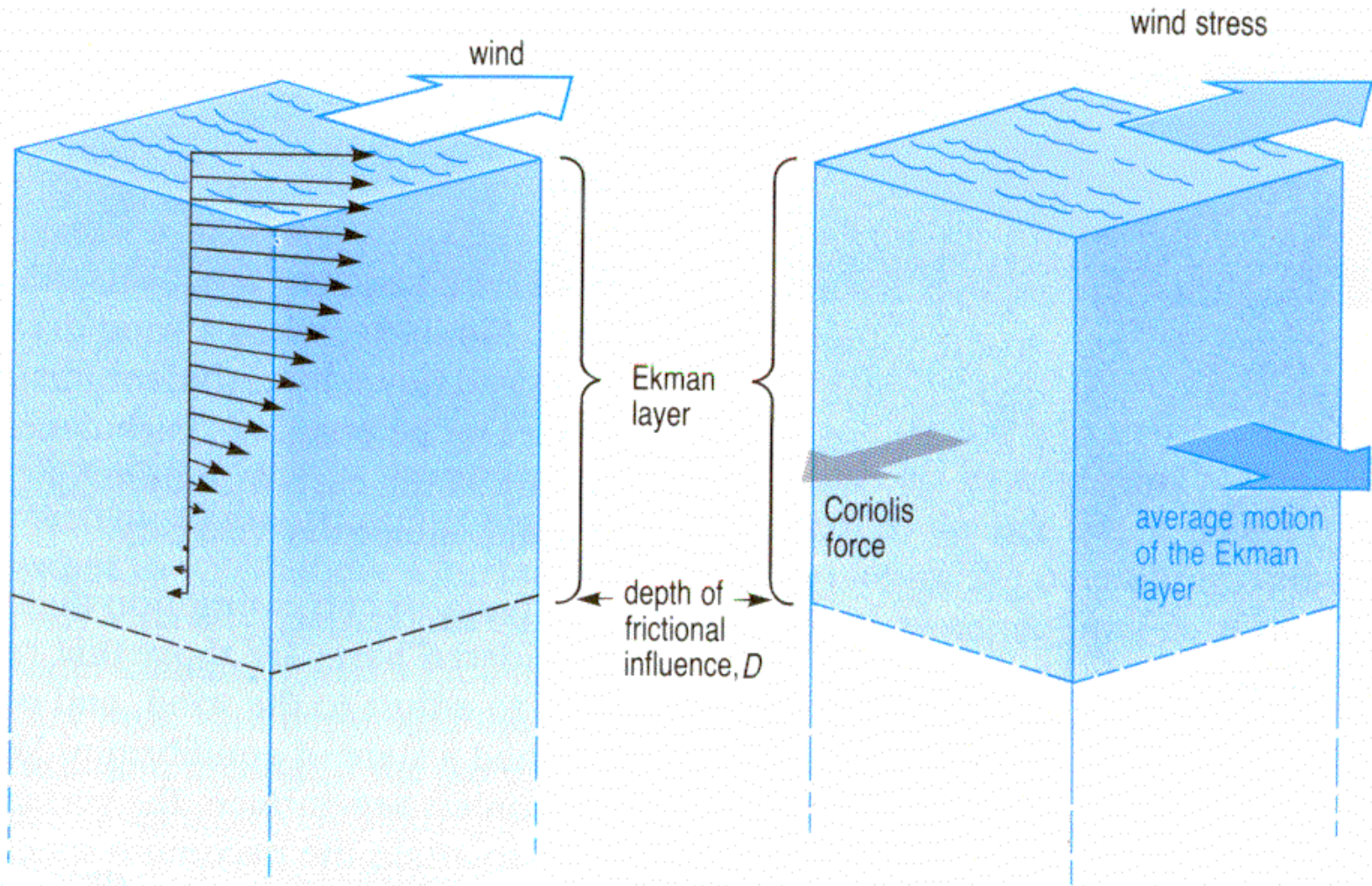
$$-f v_e = \frac{1}{\rho} \frac{\partial \tau^x}{\partial z}$$

$$f u_e = \frac{1}{\rho} \frac{\partial \tau^y}{\partial z}$$

$$U_{ek} = \frac{\tau_s^y}{\rho f}, \quad V_{ek} = -\frac{\tau_s^x}{\rho f}$$

Ekman transport right to the wind in Northern Hemisphere
and left to the wind in Southern Hemisphere

The Ekman Layer (summary)



(Northern hemisphere shown)

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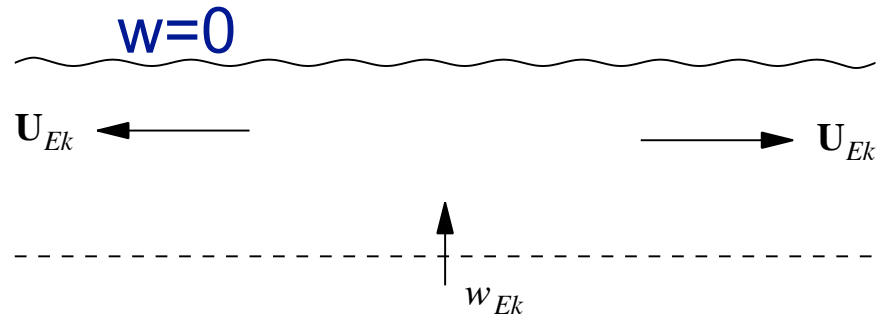
Vertical (density driven?) circulation

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How does Ekman layer influence the ocean interior?

Integrate continuity equation between $z=0$ and $z=-h$

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0 \quad \Rightarrow \quad \frac{\partial U_e}{\partial x} + \frac{\partial V_e}{\partial y} - w(-h) = 0$$



Ekman pumping velocity

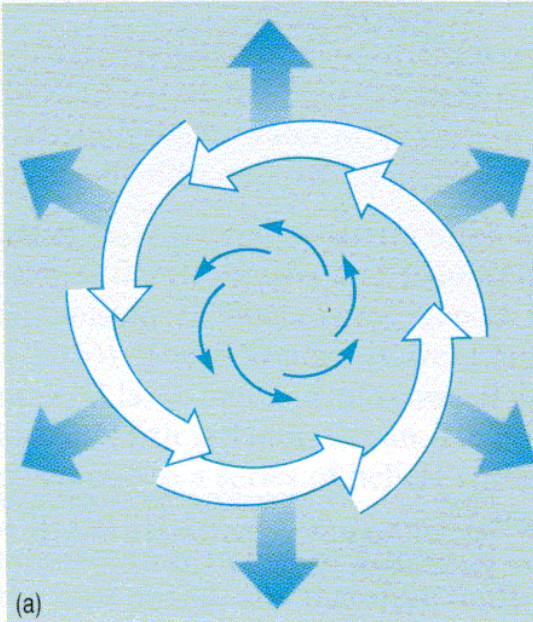
$$w(-h) = \frac{\partial}{\partial x} \left(\frac{\tau_s^y}{\rho f} \right) - \frac{\partial}{\partial y} \left(\frac{\tau_s^x}{\rho f} \right) = \frac{1}{\rho} \text{curl} \left(\frac{\tau_s}{f} \right)$$

Variations in τ (or f) induce upwelling/downwelling at basis of Ekman layer driving interior flow. Variations in τ tend to dominate except near the equator

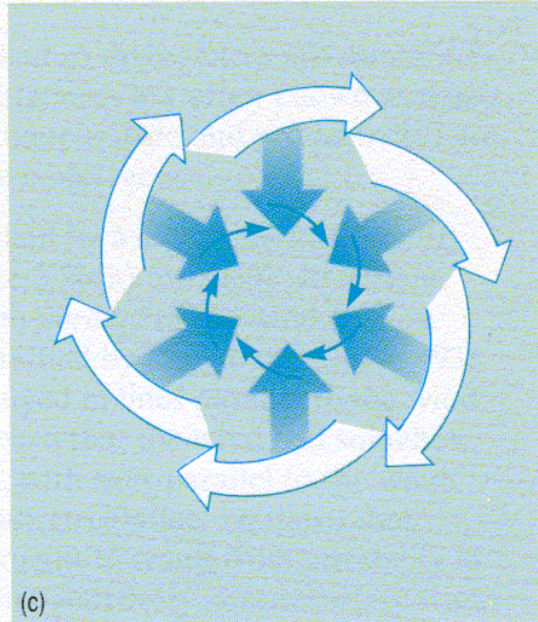
Ekman pumping examples (1)

NORTHERN HEMISPHERE

CYCLONIC WIND



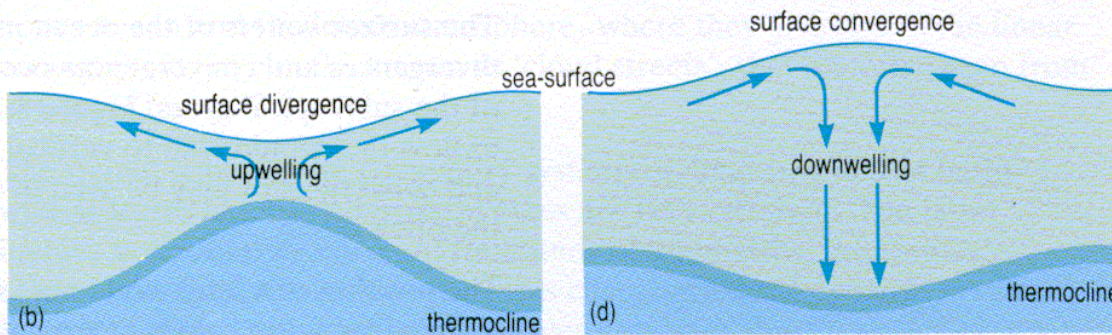
ANTICYCLONIC WIND



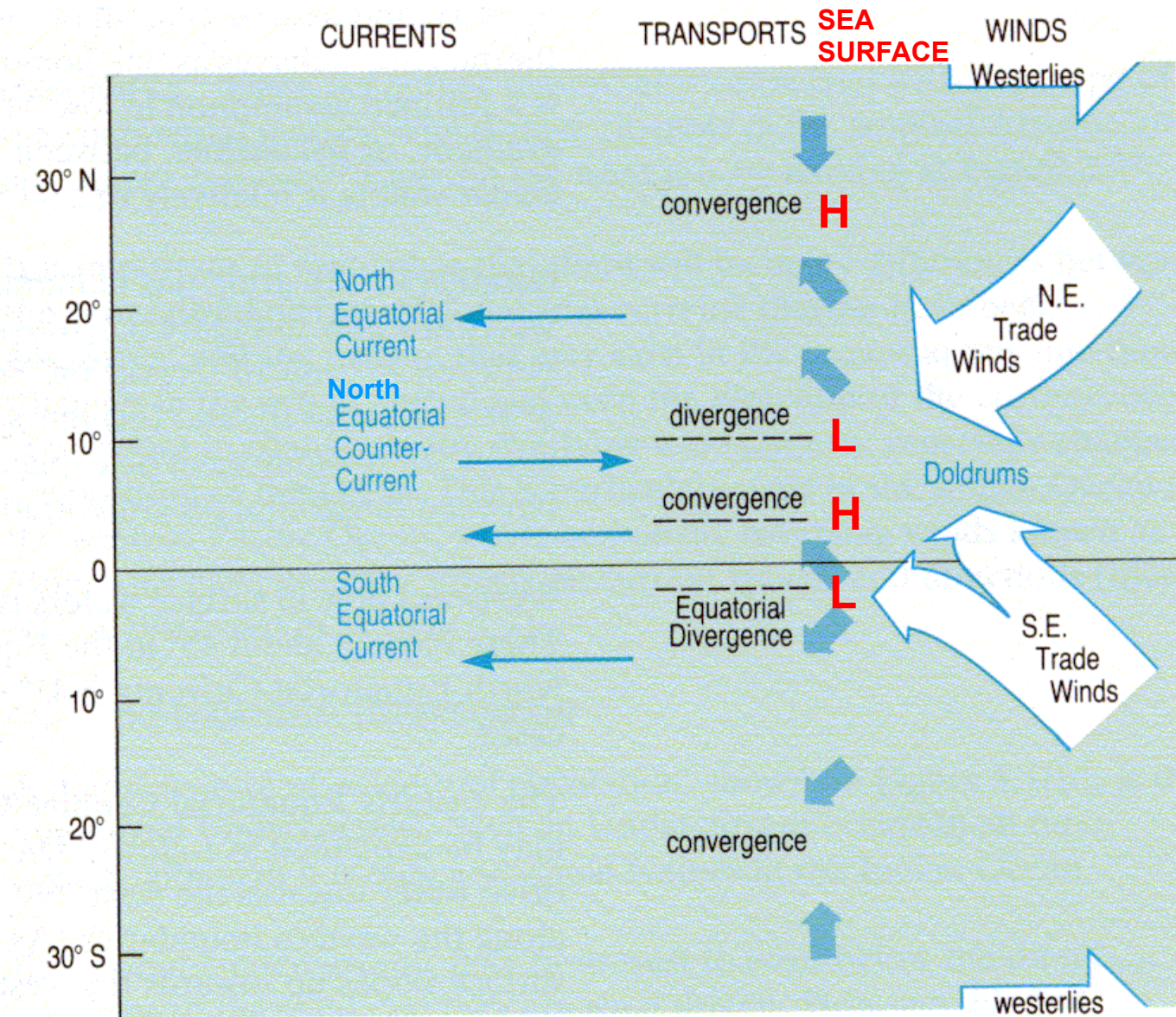
In both hemispheres:

**Cyclonic winds
produce upwelling
(Ekman suction)**

**Anticyclonic winds
produce downwelling
(Ekman pumping)**



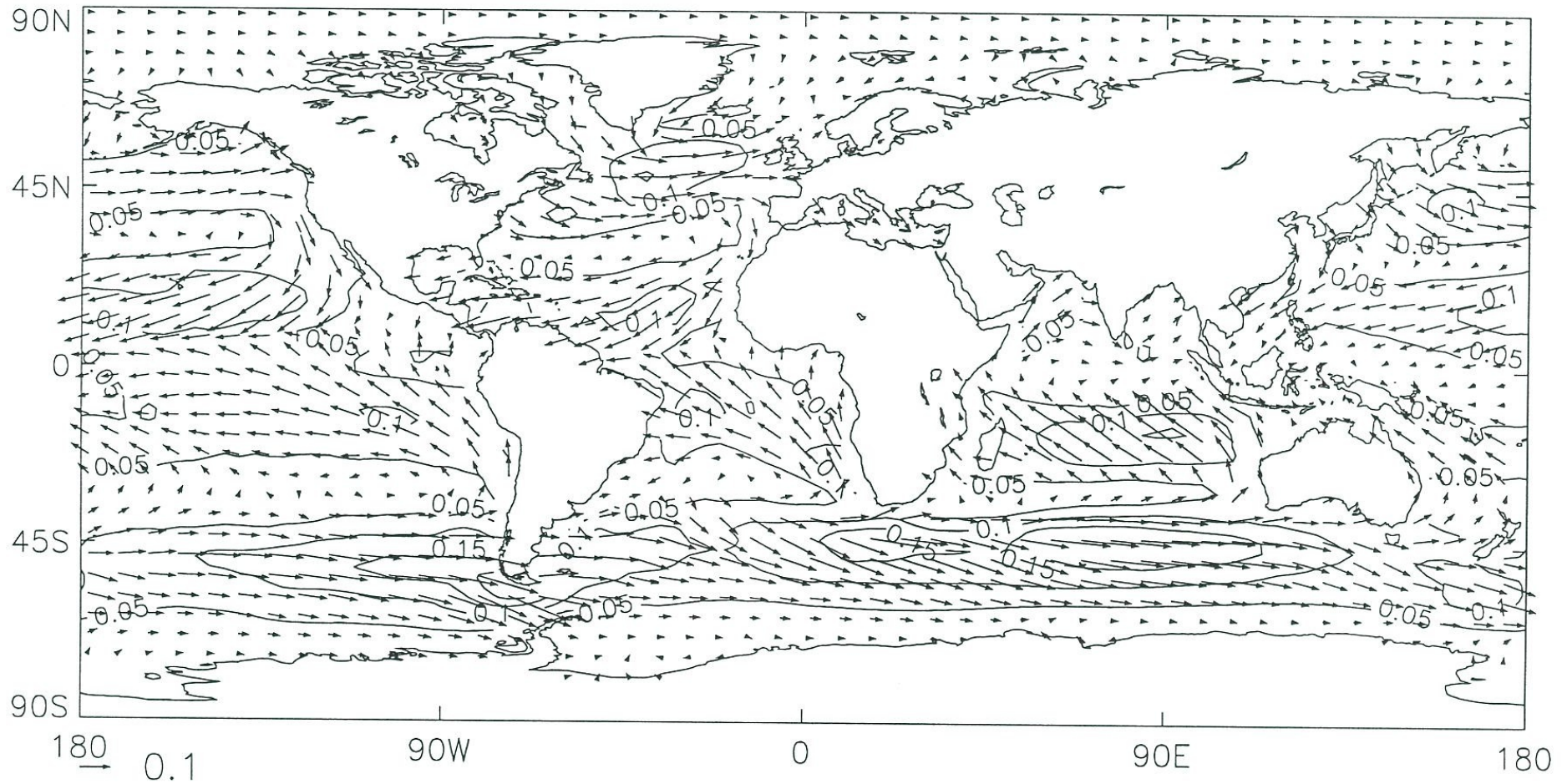
Ekman pumping examples (2): Tropical Currents



Exercise: find a wind stress climatology, and calculate approximately where the Ekman pumping is dominated by variations in f , and where by variations in τ

Observed wind stress

Wind Stress (Nm^{-2})
Hellerman & Rosenstein climatology. Annual mean



Exercise: To what extent can you use this to explain the sea surface height variations shown at the beginning?

← polar easterlies

→ midlatitude
westerlies

← trades



$$w_{Ek} > 0$$

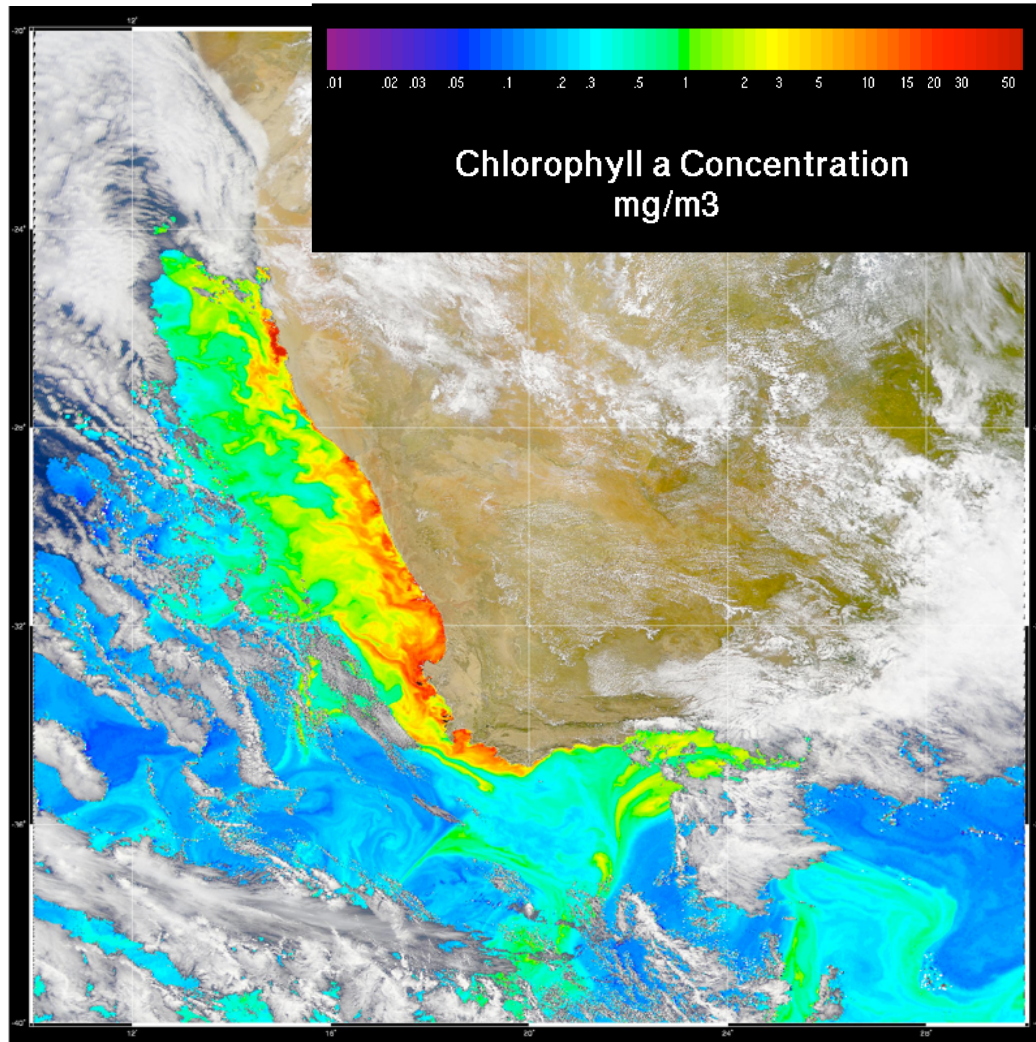


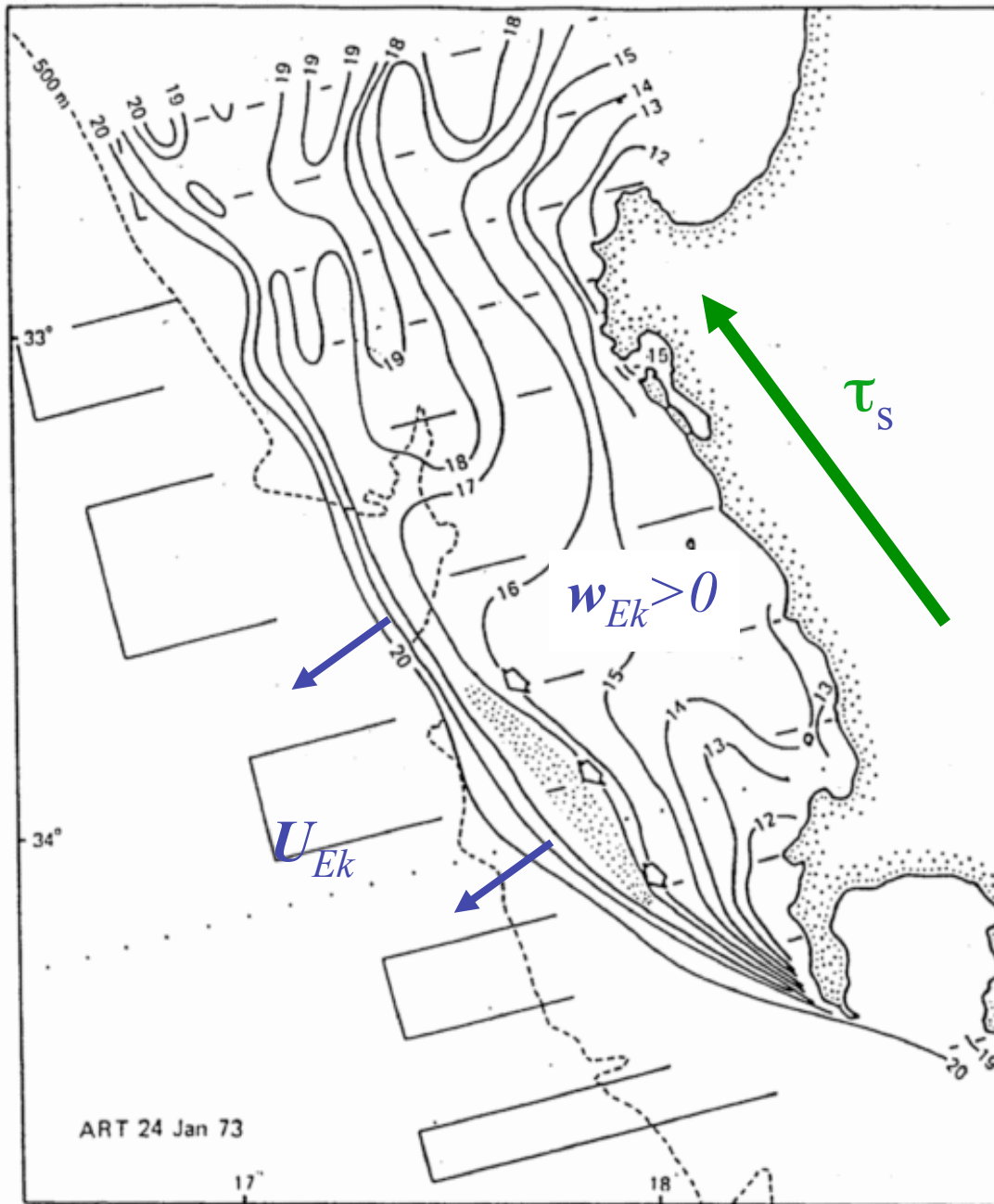
$$w_{Ek} < 0$$



$$V_{Ek}$$

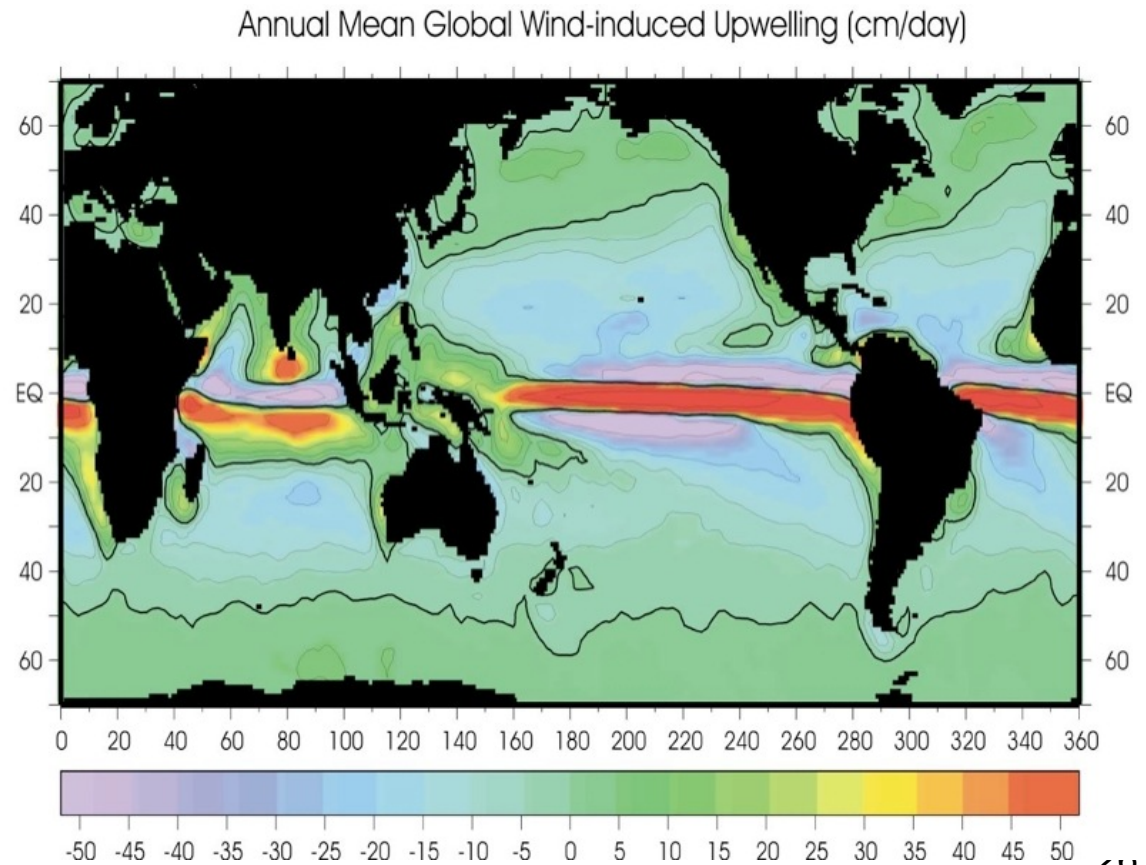
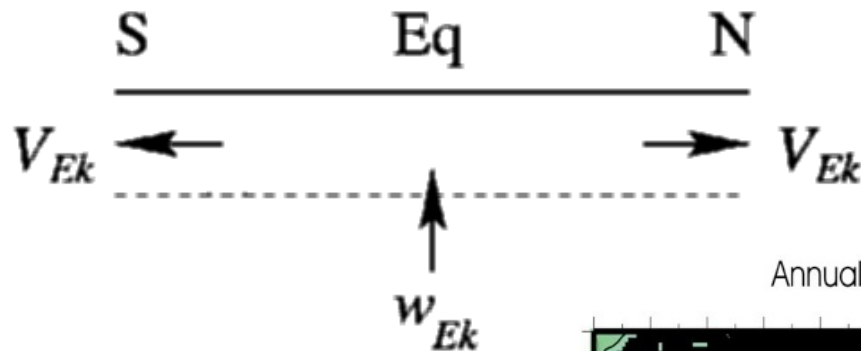
Chlorophyll a concentration near South Africa/Namibia





Theory
explains
nutrient rich
regions
observed
near many
coastal areas
➡ coastal
upwelling

Theory explains equatorial upwelling

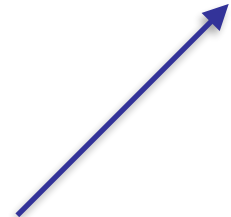


Near equator, variations in f may sometimes dominate
=> equatorial downwelling

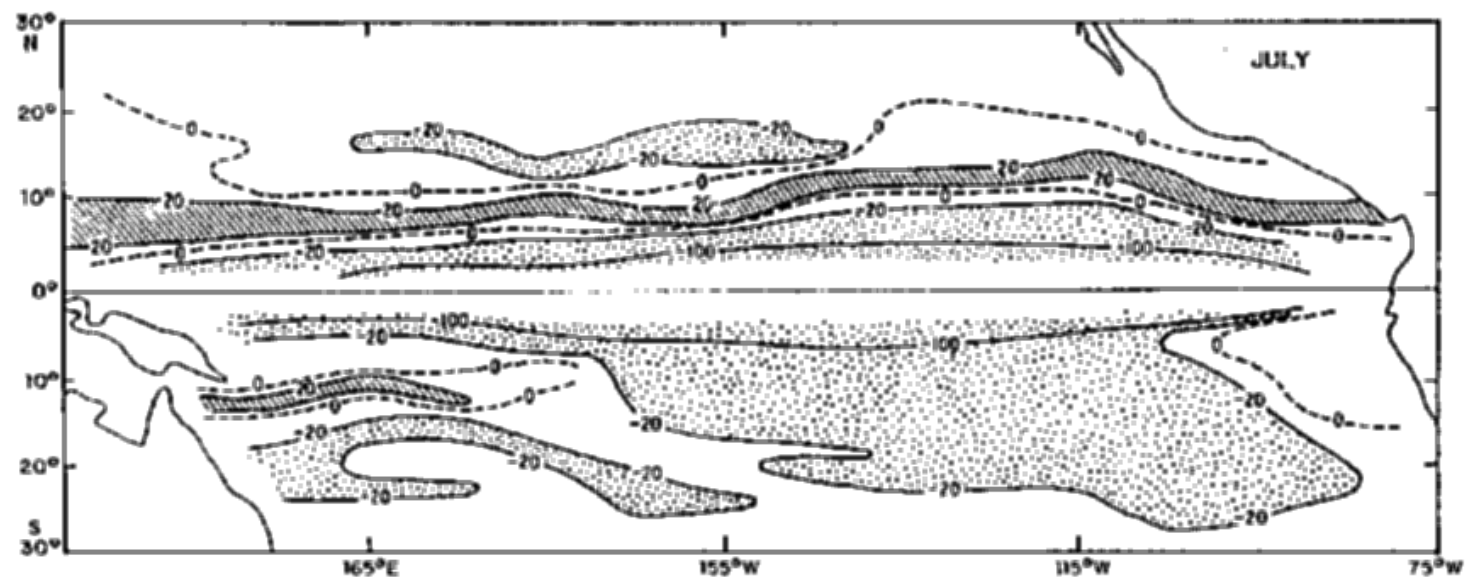
$$w_e = -\frac{1}{\rho} \frac{\partial}{\partial y} \left(\frac{\tau_x}{f} \right) = -\frac{1}{\rho f} \frac{\partial \tau_x}{\partial y} + \frac{\beta \tau_x}{\rho_0 f^2}$$



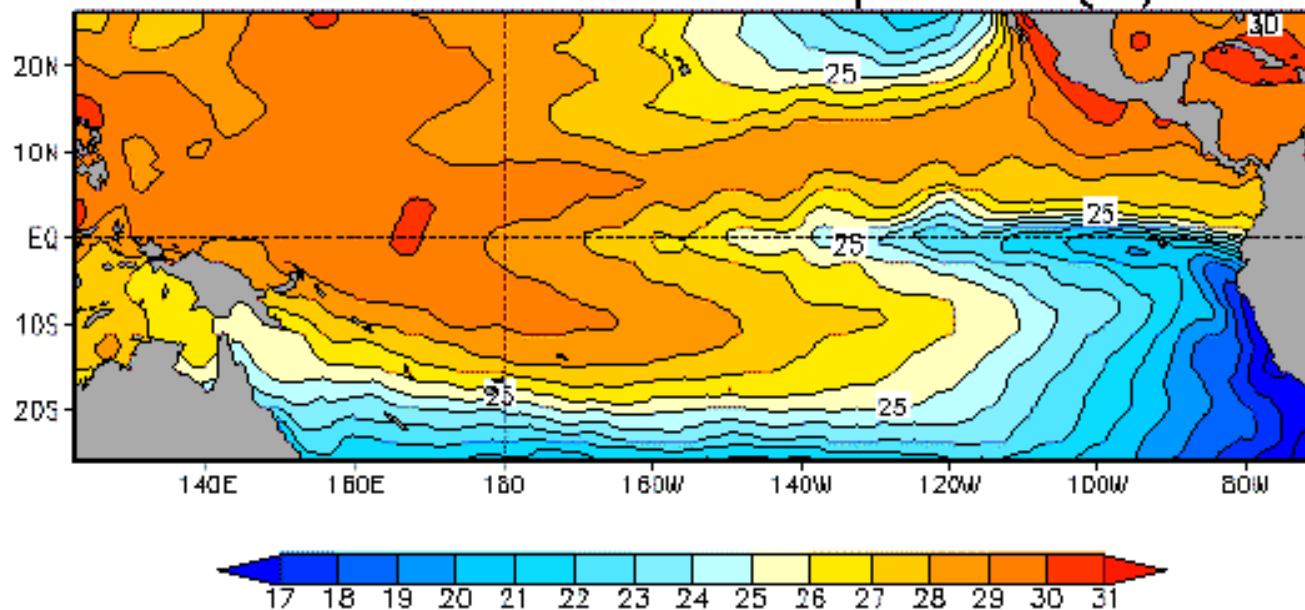
Neglect contribution
from meridional
wind stress



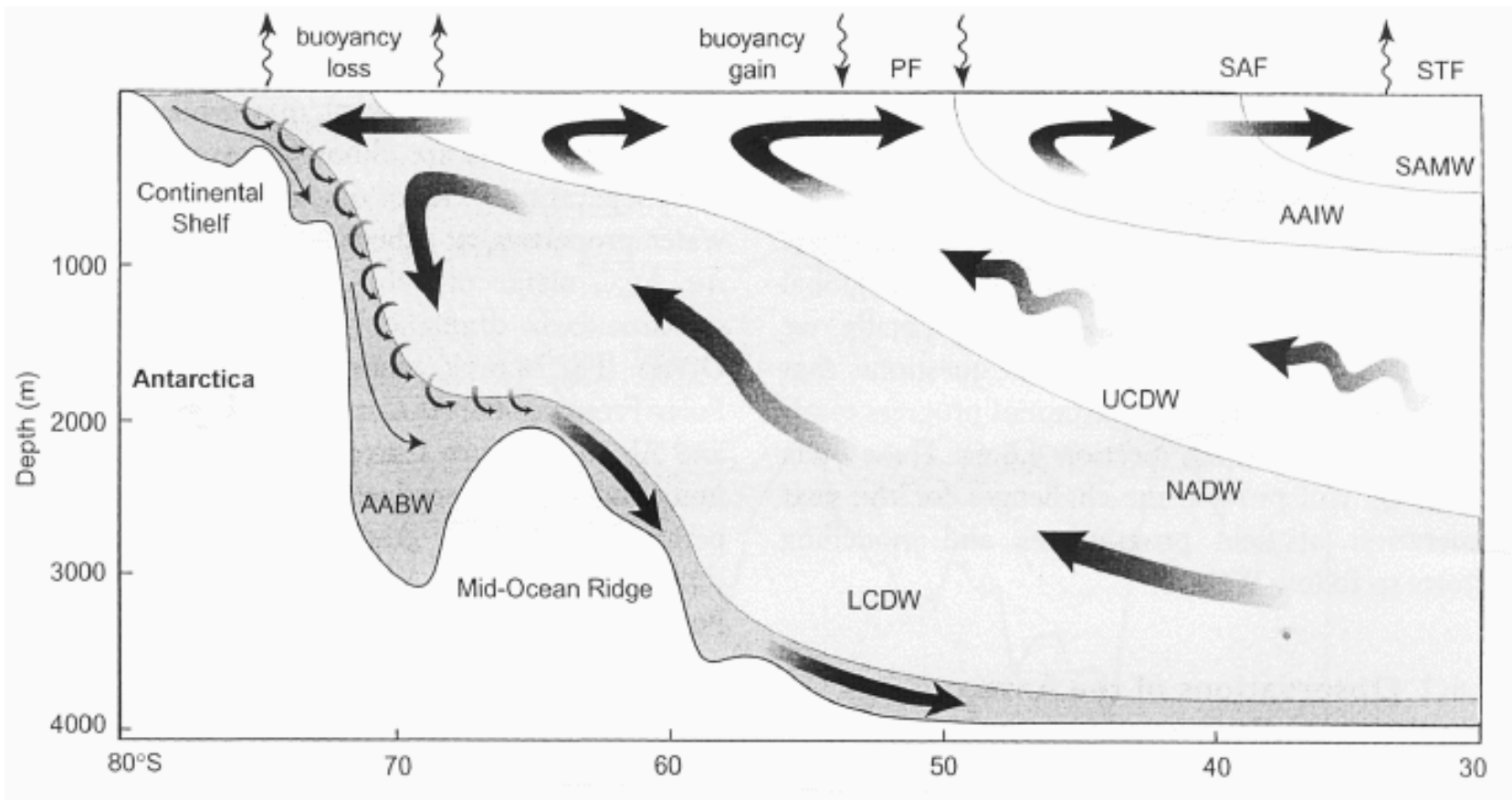
Trade winds => Negative



Observed Sea Surface Temperature ($^{\circ}\text{C}$)



The Antarctic Circumpolar Ocean as a big coastal upwelling



Contributes to the tilting of isopycnal surfaces, and by the thermal wind relationship, to maintaining vertical shear and the ACC

Quantifying the effects of Ekman pumping on the interior: Sverdrup transport

Away from boundaries and on the scale of ocean basins, ocean is approximately geostrophic (scaling left as exercise for reader!)

Consider steady, geostrophic flow:

- (1) $-fv = -p_x/\rho$ [p_x means $\partial p/\partial x$, etc]
- (2) $fu = -p_y/\rho$
- (3) $u_x + v_y + w_z = 0$

Eliminate p from (1) and (2) by cross-differentiating:

$$fu_x + f v_y + \beta v = 0 \quad \text{[where } \beta = \partial f/\partial y \text{]}$$

Using (3) gives:

(4) $\beta v = fw_z$

[Linear vorticity balance or geostrophic ballerina: stretch a water column and it moves poleward so it is spinning faster]

Sverdrup transport (continued)

Integrate (4) from base of Ekman layer down to a 'level of no motion' (or flat bottom) where $w=0$:

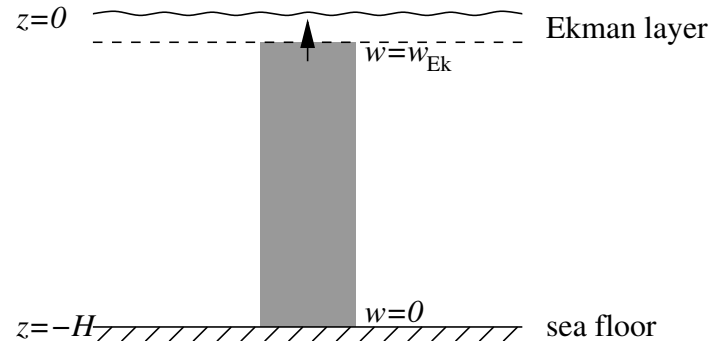
$$\beta V = f(w_E - w_{\text{bot}})$$

where V is the depth-integrated transport in the interior layer

$$V = \int v \, dz .$$

So

- $\beta V = fw_E = (f/\rho) \text{ curl}(\underline{\tau}/f)$

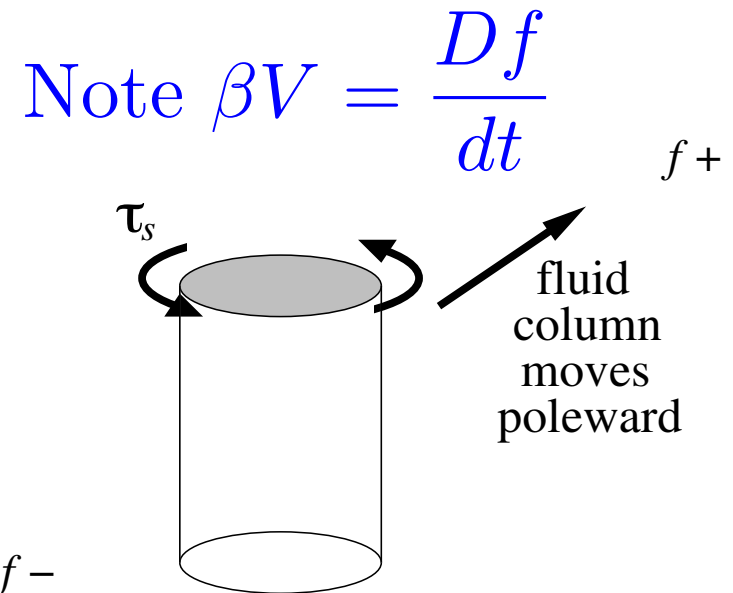
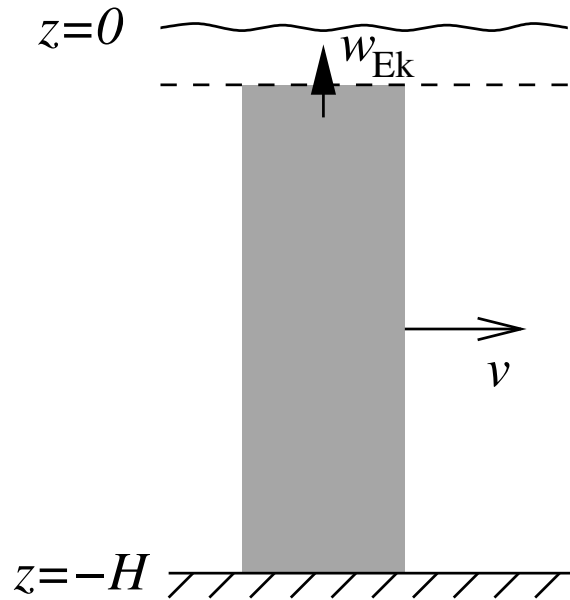


This is the Sverdrup balance. The wind stress curl sets the gyre strength (at least in this theory). In regions of *anticyclonic* wind stress curl, the integrated geostrophic interior flow is *toward* the equator (in both hemispheres).

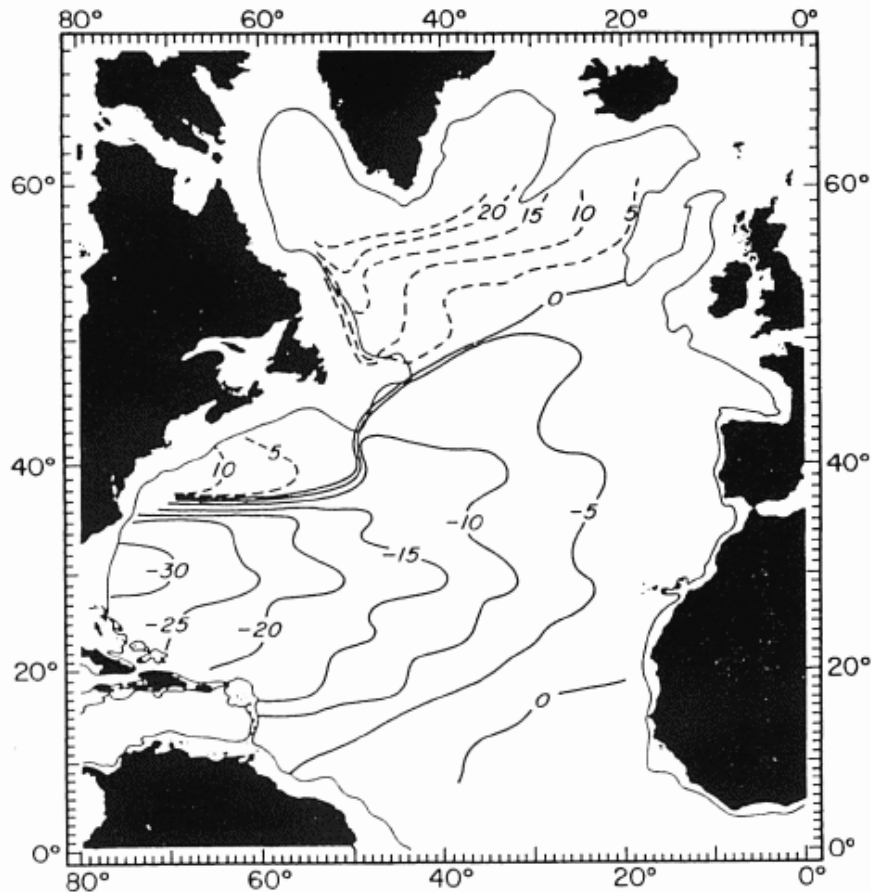
Sverdrup balance $\beta V = fw_e$

upwelling (positive curl) => poleward motion

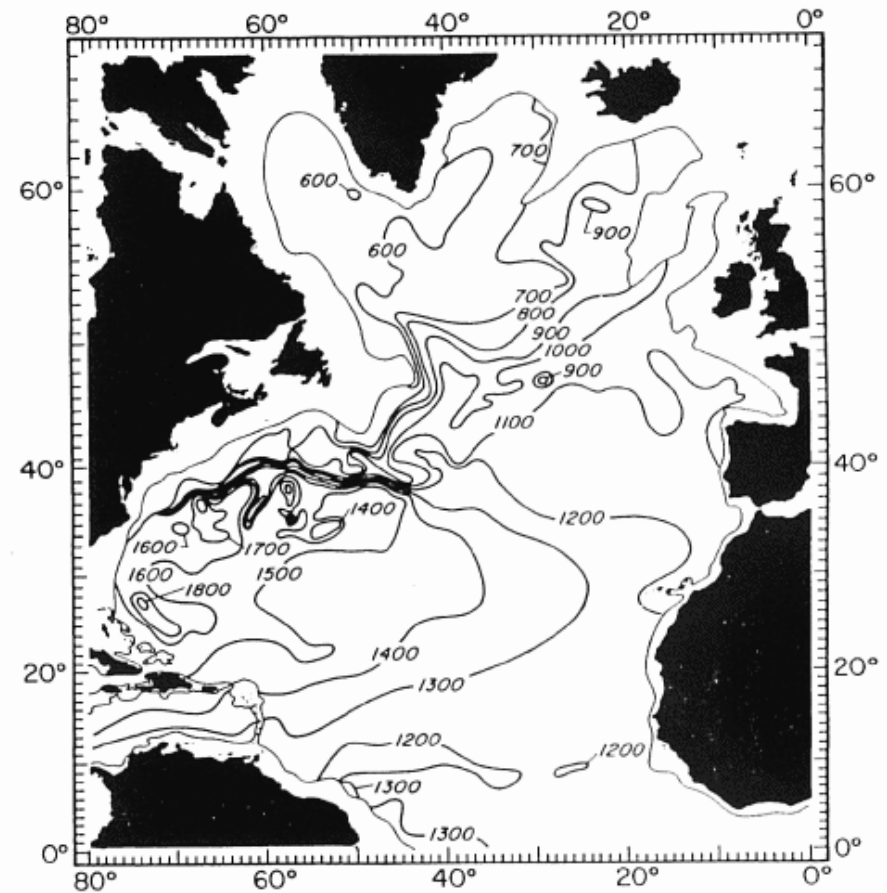
downwelling (negative curl) => southward motion



Geostrophic transport from Sverdrup balance



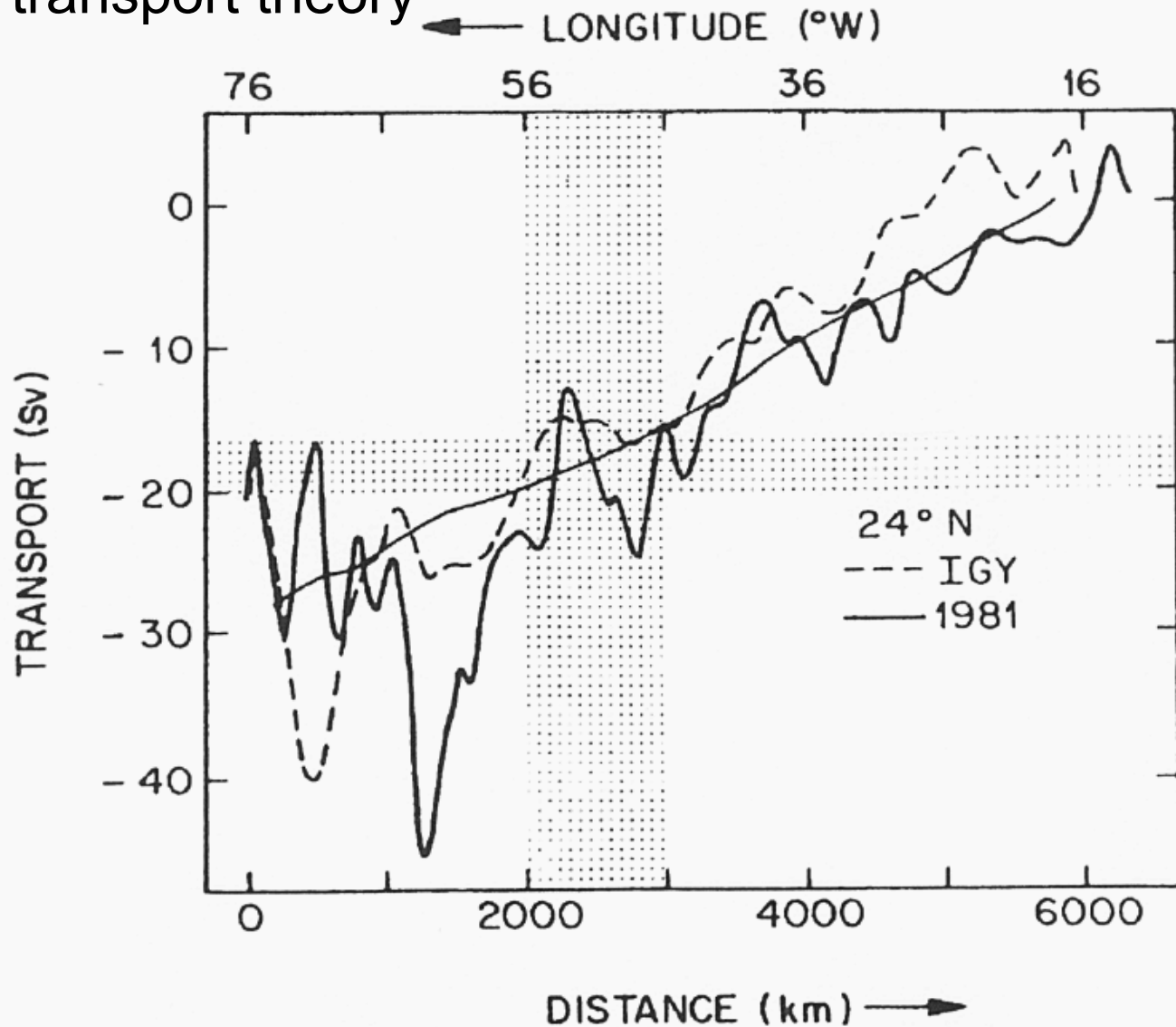
Dynamic topography at 100m



Gill (1982)

Observational test of Sverdrup transport theory

Schmitz et al., 1992



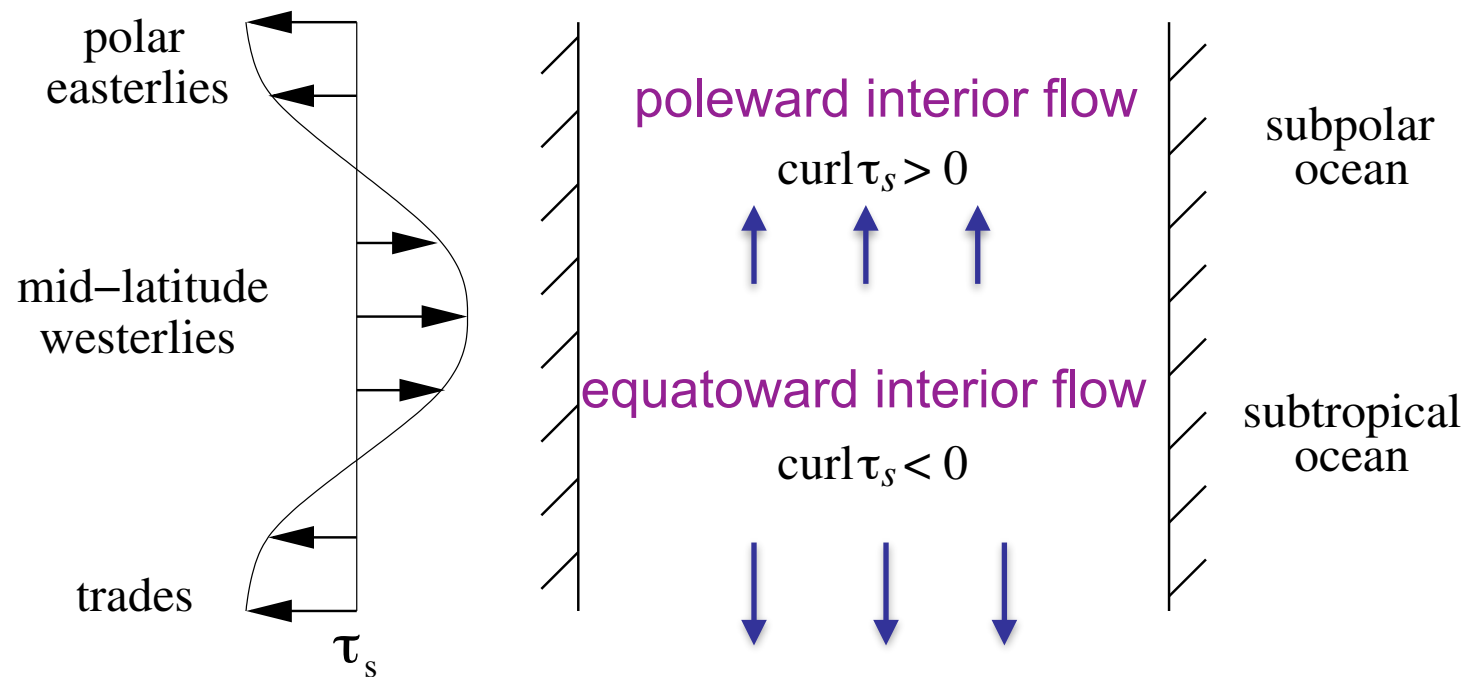
Outline

Horizontal (wind-driven?) circulation

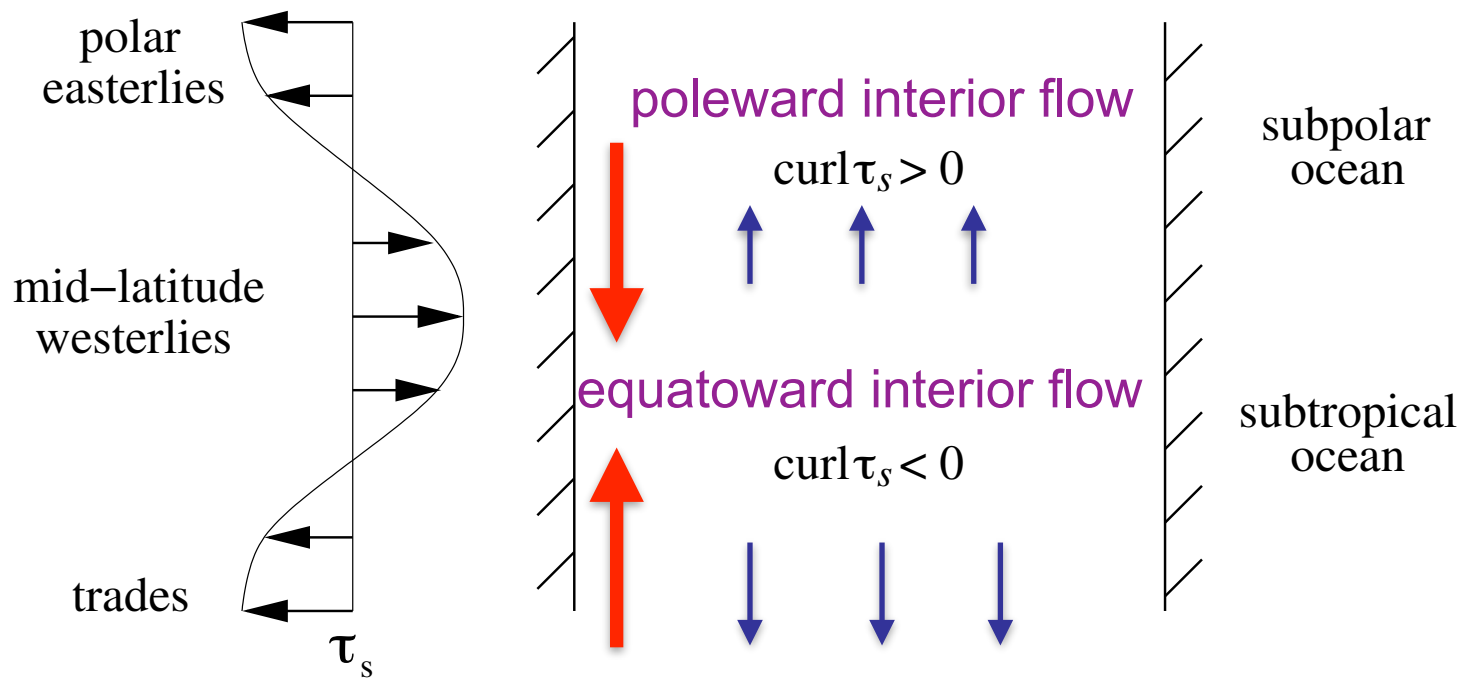
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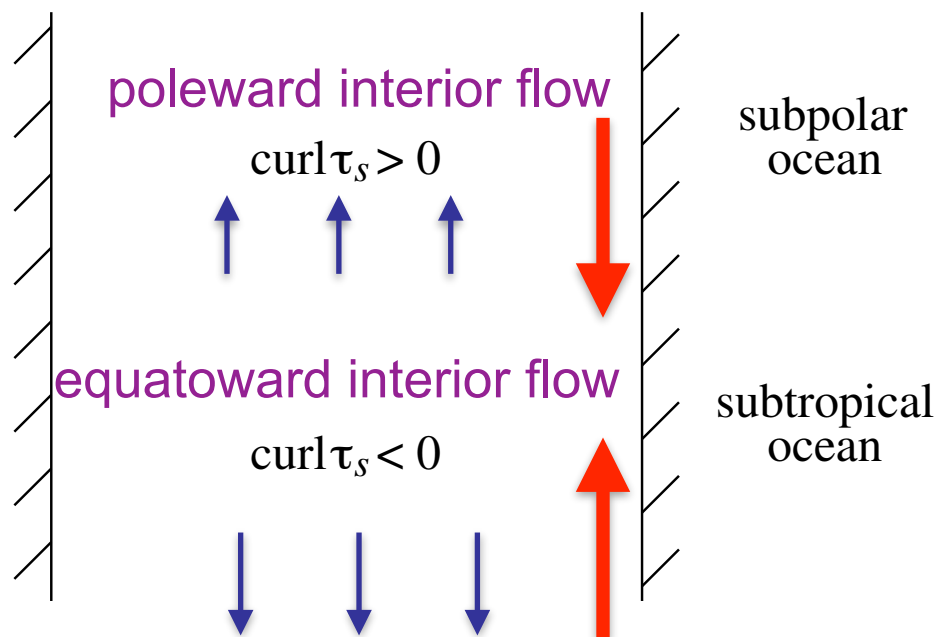
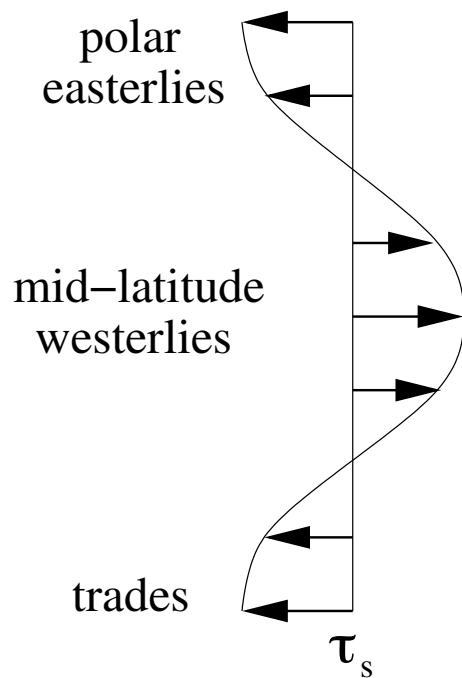
Vertical (density driven?) circulation

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- Digression: the equation of state
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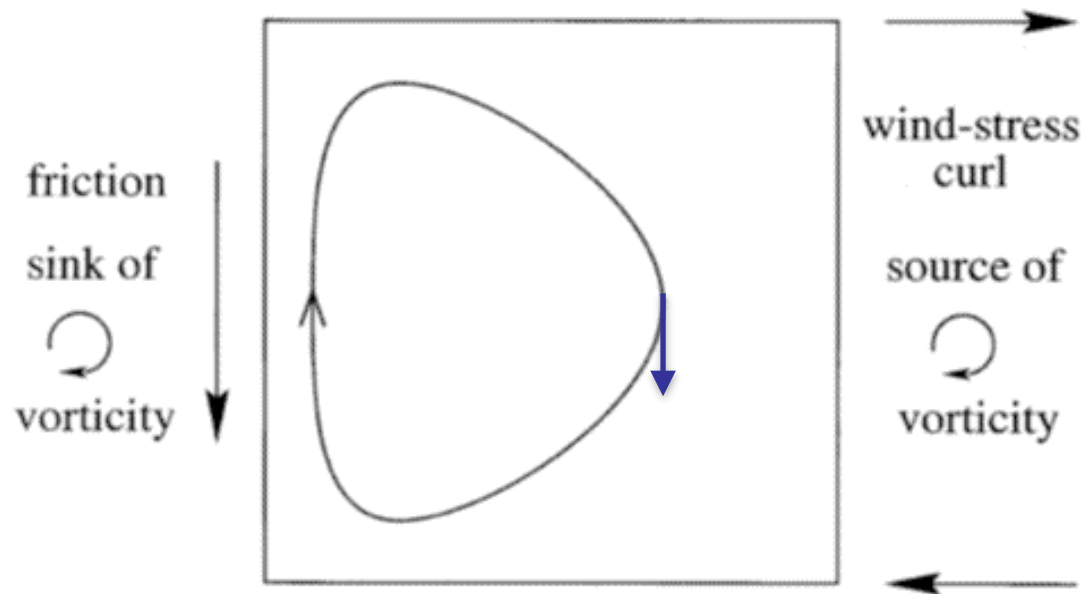
Mass conservation requires return flow somewhere
where geostrophic balance does not hold





Insights from Vorticity budget

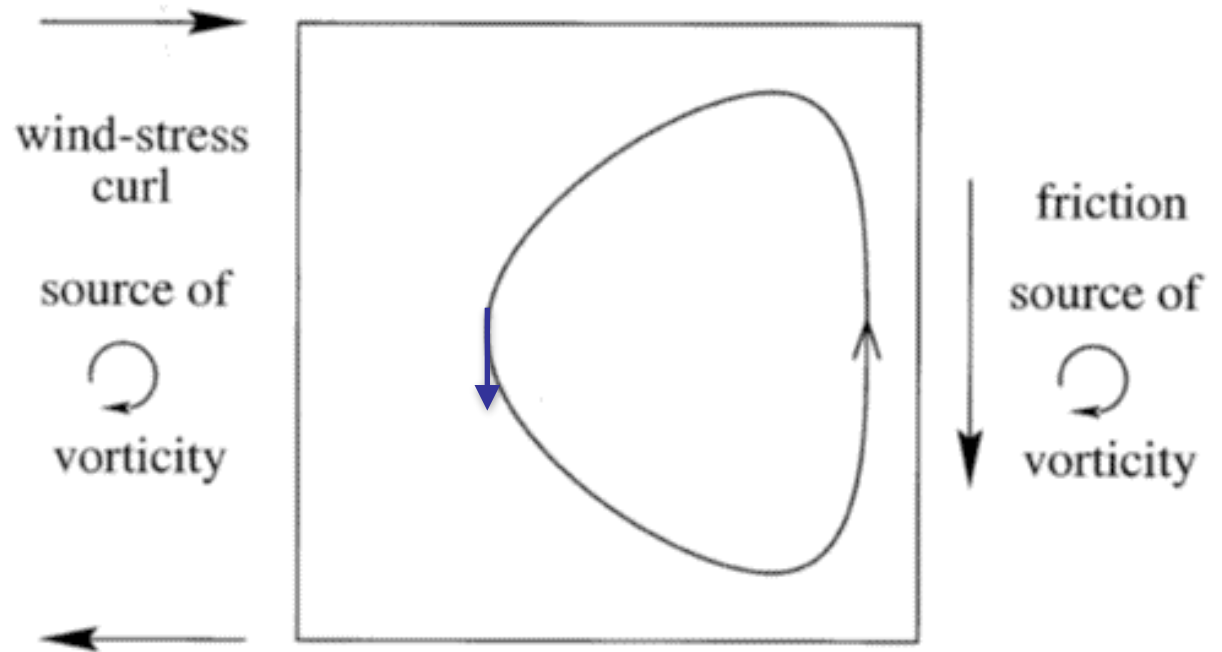
a. western boundary current:



source and sink of anticyclonic vorticity can balance

Insights from Vorticity budget (ct'd)

b. eastern boundary current:



net source of anticyclonic vorticity

**Steady State
Impossible!**

The western boundary current: the final piece of the puzzle

Example: Northern hemisphere subtropical gyres – see sea surface height observations

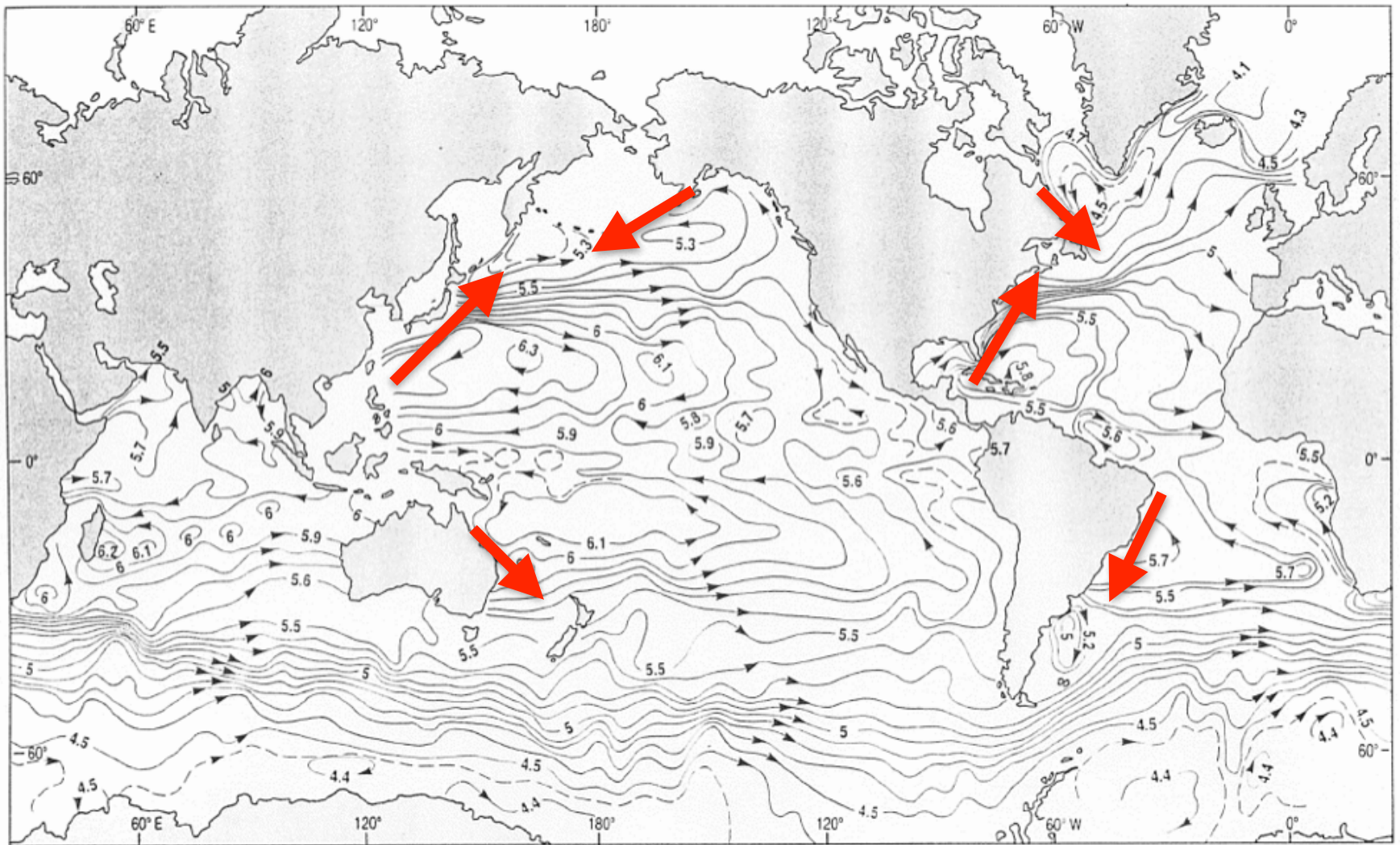
Sverdrup transport says that interior flow is southward since $\text{curl } \underline{\tau}$ anticyclonic. But flow can't be southward everywhere or the North Atlantic would empty out. Need a northward return flow somewhere to balance it – somewhere where geostrophy is violated.

Various possibilities, e.g.

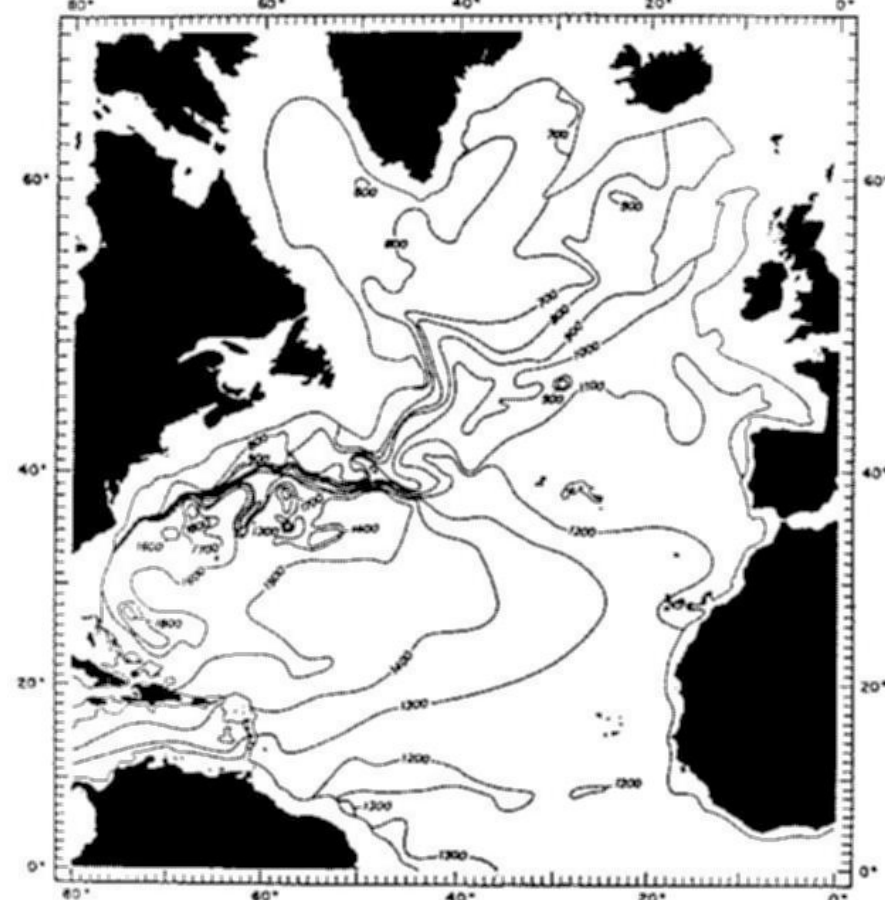
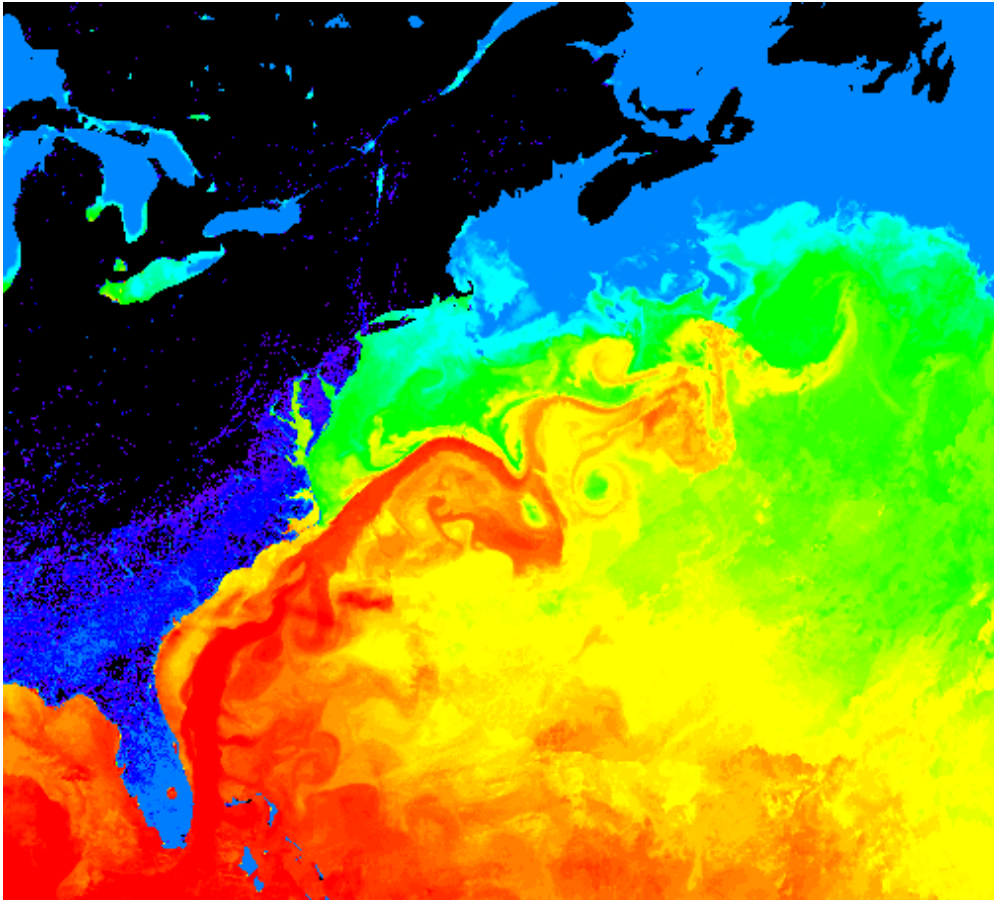
- i. In the Ekman layer**
- ii. In a frictional boundary layer on the eastern side**
- iii. In a frictional boundary layer on the western side**

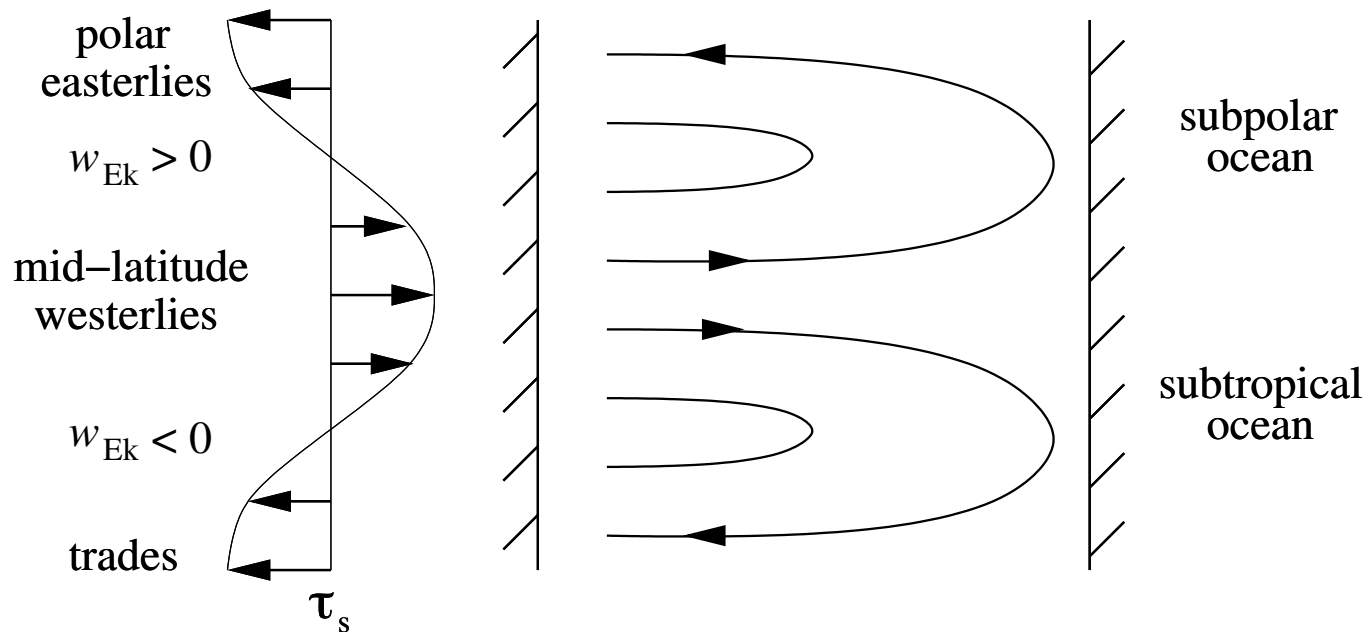
Exercise:

By considering the vorticity budget of a fluid column as it travels round the gyre, convince yourself that a northward flowing boundary layer can only complete the gyre if it is on the *west* of the basin. This is what is observed (e.g. see SSH map). See Pond and Pickard for argument.



Observed Western Boundary Currents





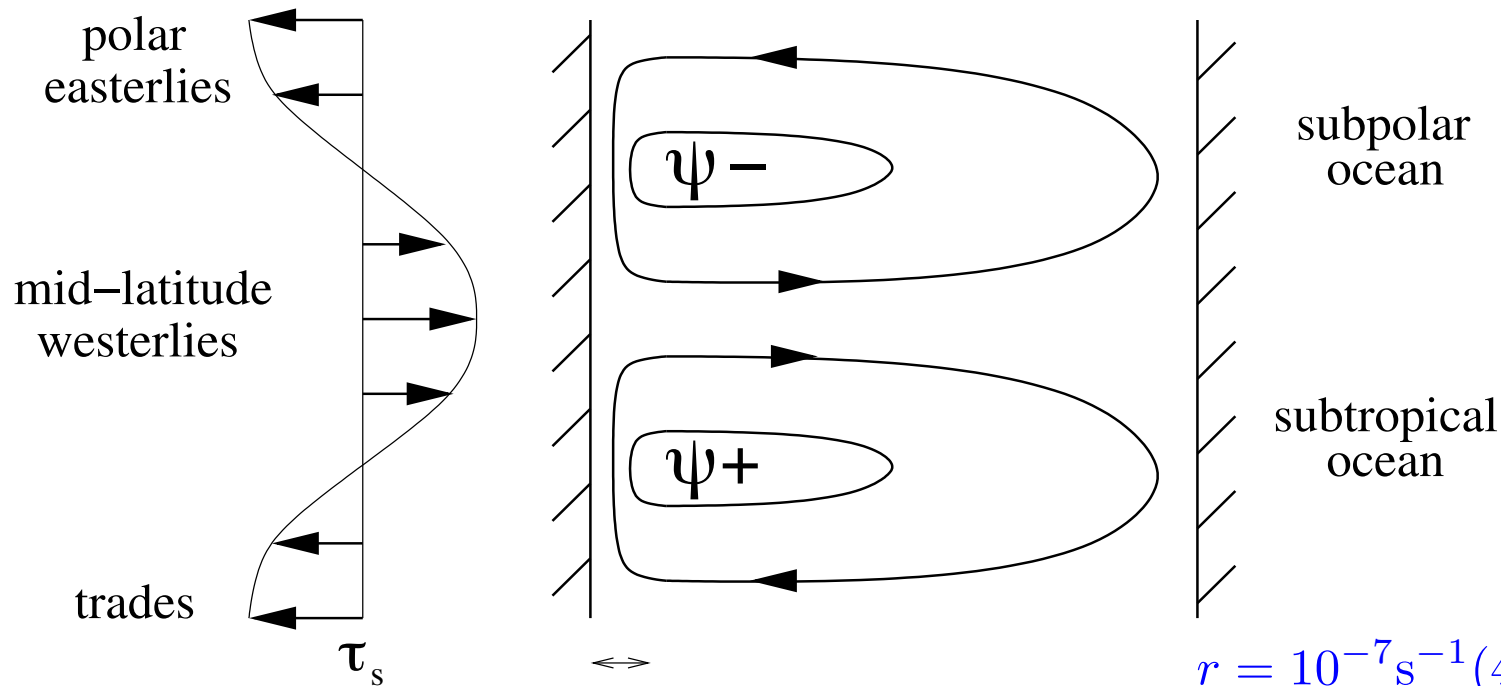
How can we analytically close the mass budget in the western boundary currents?

Henry “Hank” Stommel



Stommel model Analytic closure of western boundary

$$\begin{aligned} -fv + \frac{1}{\rho} \frac{\partial p}{\partial x} &= -ru \\ fu + \frac{1}{\rho} \frac{\partial p}{\partial y} &= -rv \end{aligned} \quad \begin{array}{l} \nwarrow \\ \swarrow \end{array} \quad \begin{array}{l} \text{Bottom} \\ \text{Friction} \end{array}$$



$$\delta_{stommel} = \frac{r}{\beta}$$

$$r = 10^{-7} \text{s}^{-1} (4 \text{ months})$$

$$\beta = 2.10^{-11} \text{m}^{-1} \cdot \text{s}^{-1}$$

$$\delta_{stommel} = \frac{10^{-7}}{2.10^{-11}} = 5 \text{ km!!}$$

Modelling the western boundary layer

Redo Sverdrup calculation, but with the addition of a lateral eddy viscosity ('friction') term:

$$-fv = -p_x/\rho + Au_{xx} \quad [A = \text{eddy viscosity}]$$

$$fu = -p_y/\rho + Av_{xx}$$

$$u_x + v_y + w_z = 0$$

Same procedure as before gives:

$$\beta v = fw_z + A(v_x - u_y)_{xx} \quad [\text{assume } u_y \ll v_x \text{ in western BL}]$$

Sverdrup balance becomes:

$$\beta V = fw_E + AV_{xxx}$$

Third term small in interior. Only becomes important in a narrow boundary layer of width L .

Modelling the western boundary layer (2)

Viscous term becomes important when AV_{xxx} is the same size as βV , i.e.

$$\beta V \sim AV/L^3$$

or
$$L \sim (A/\beta)^{1/3}$$

E.g. for HadCM3, A is around $4500 \text{ m}^2\text{s}^{-1}$ at midlatitudes. Gives $L \sim 60\text{km}$, just about resolved by model grid (1.25°).

More generally, in a GCM need to resolve the western boundary layer, so need $\Delta x < (A/\beta)^{1/3}$, or

$$A > \beta \Delta x^3.$$

The cube dependence on Δx means that coarse resolution models need very high A in order to have a resolved boundary layer.

In such models A is chosen for reasons of numerical stability – too low a value gives spurious oscillatory solutions or numerical blowups.

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Vertical circulation

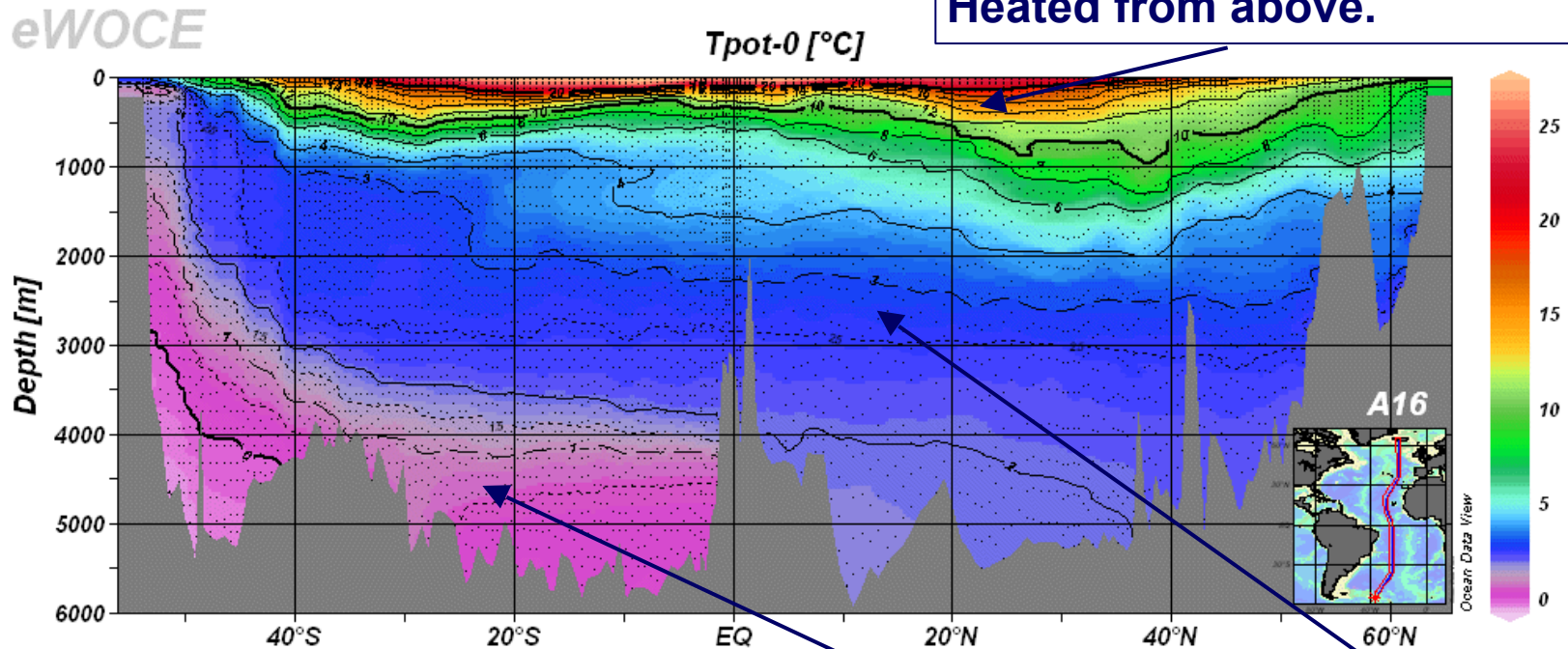
While horizontal circulation tends to be thought of as wind-driven, and has a well-developed theory, the vertical ('overturning') circulation tends to be thought of as density-driven, and has a less well developed theory.

(All the above is true to leading order only)

Potential Temperature Cross Section: Atlantic

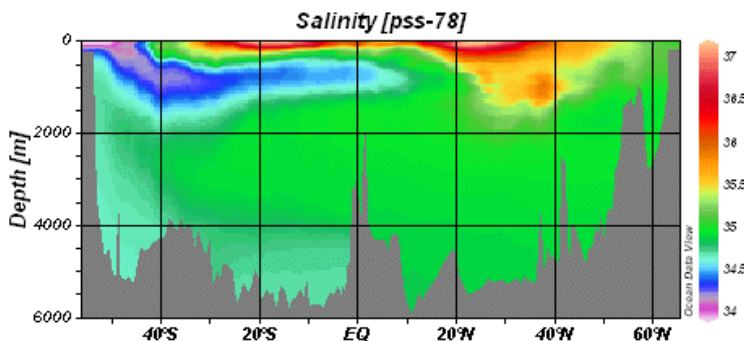
Interior heat sources are weak. So temperature is set at surface (plus effects of weak interior mixing)

Warm 'thermocline' water.
Heated from above.



Cold bottom
water ($<1^{\circ}\text{C}$).
Antarctic origin.

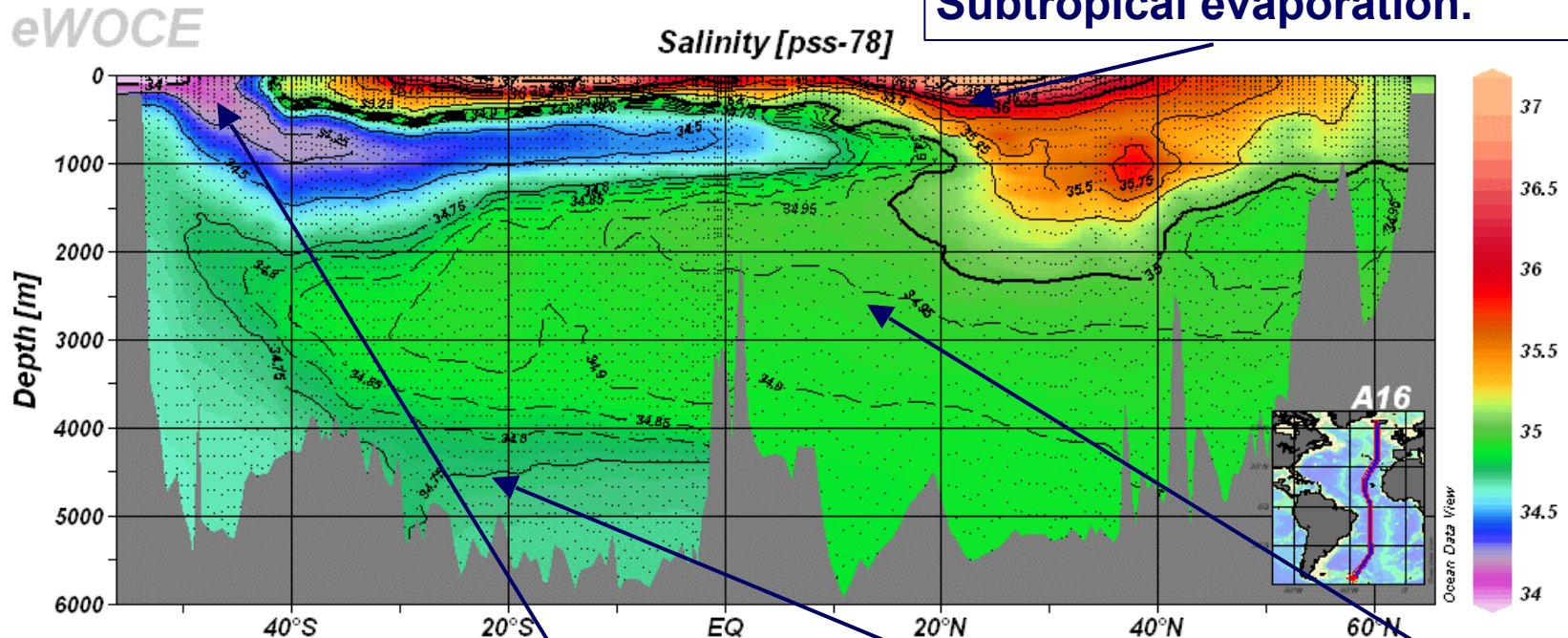
Coldish deep
water ($1-5^{\circ}\text{C}$).
Southern or
northern origin?



Salinity Cross Section: Atlantic

Interior fresh water sources are weak. So salinity is set at surface (plus effects of weak interior mixing)

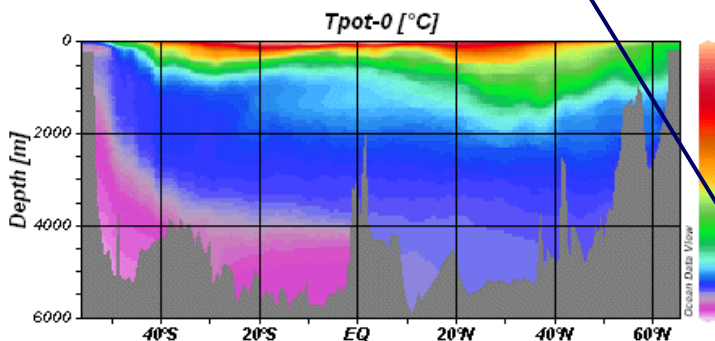
Salty 'thermocline' water.
Subtropical evaporation.



Fresh bottom water
(<34.8 psu).
Antarctic origin.

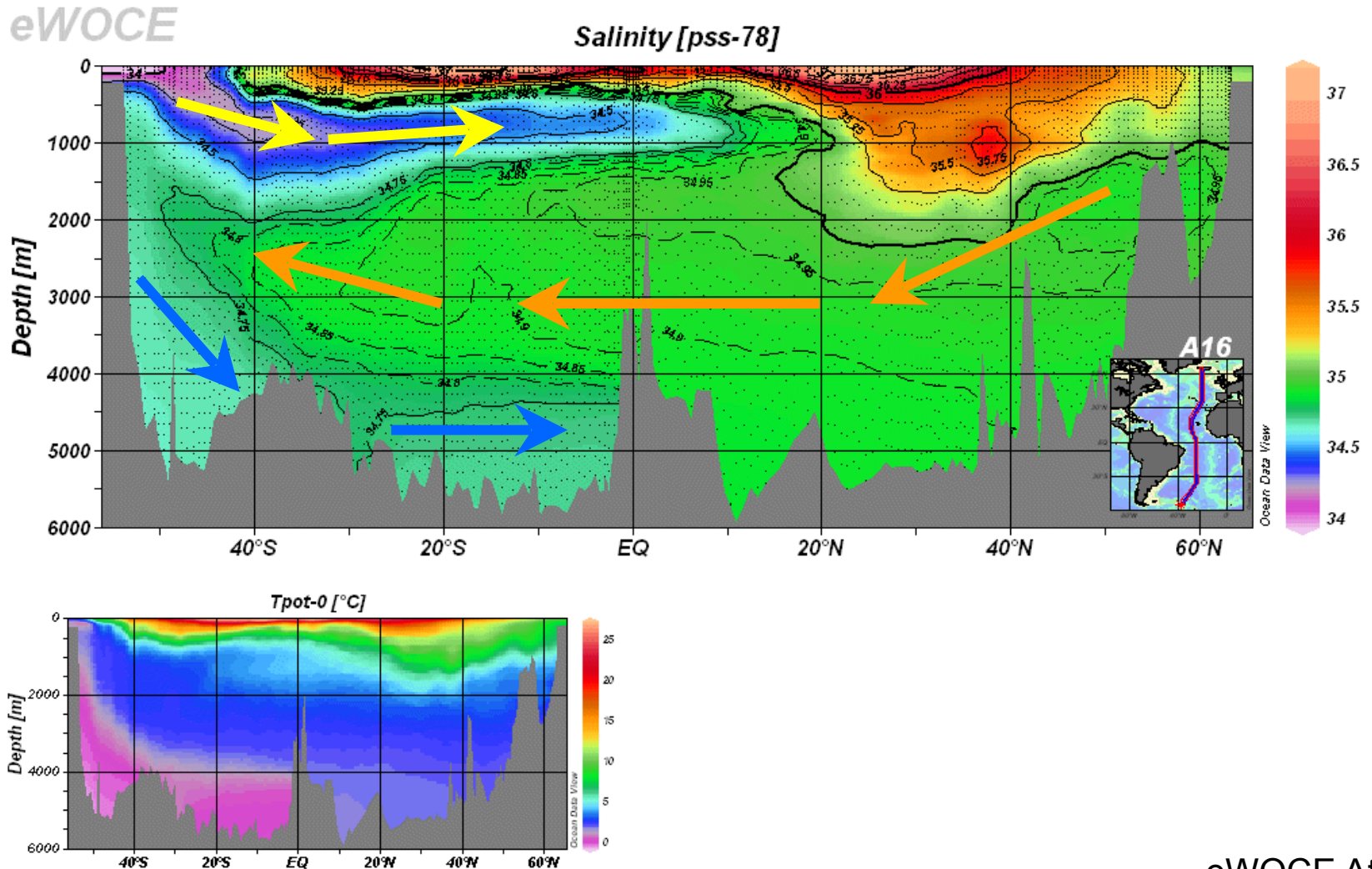
Saltyish deep water
(~ 34.9 psu).
Northern origin.

Fresh intermediate water
(<34.5 psu). Southern origin

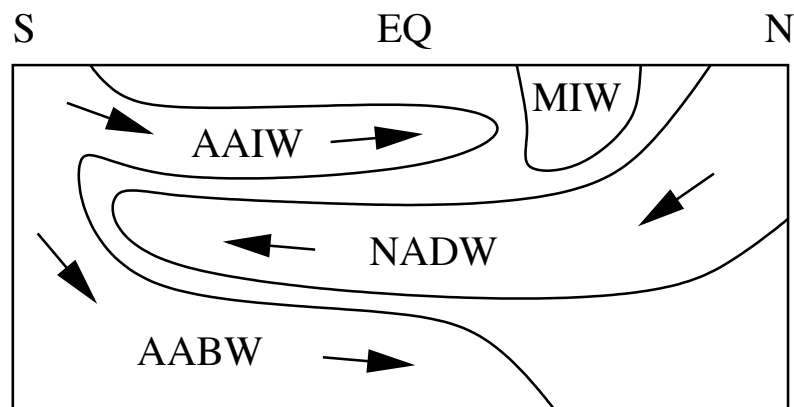
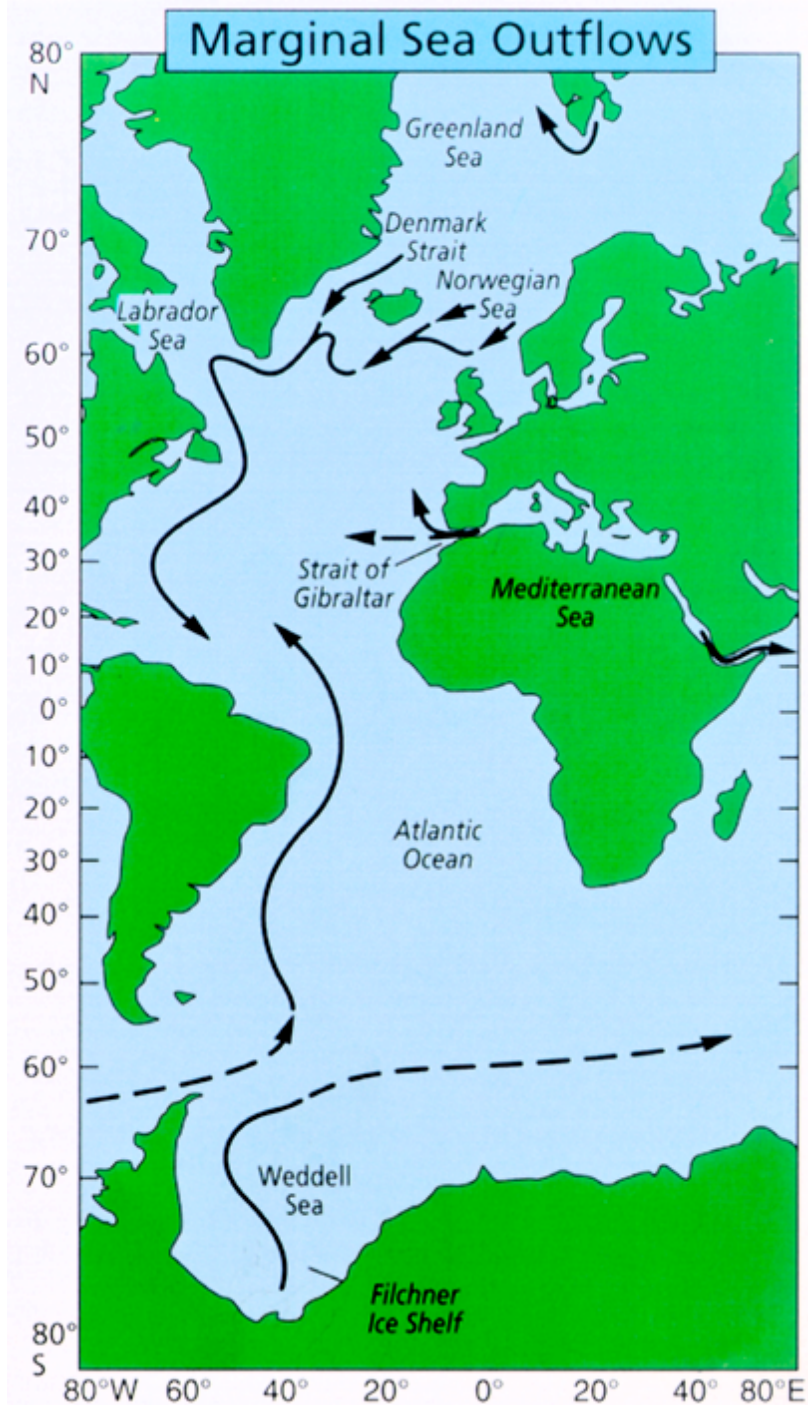


Inferences about the large scale flow

Picture of large scale water masses, their properties set initially near the surface by interaction with the atmosphere, then slowly modified by interior mixing)



Marginal Sea Outflows



Deep Water Formation in Marginal Seas

OPEN
OCEAN

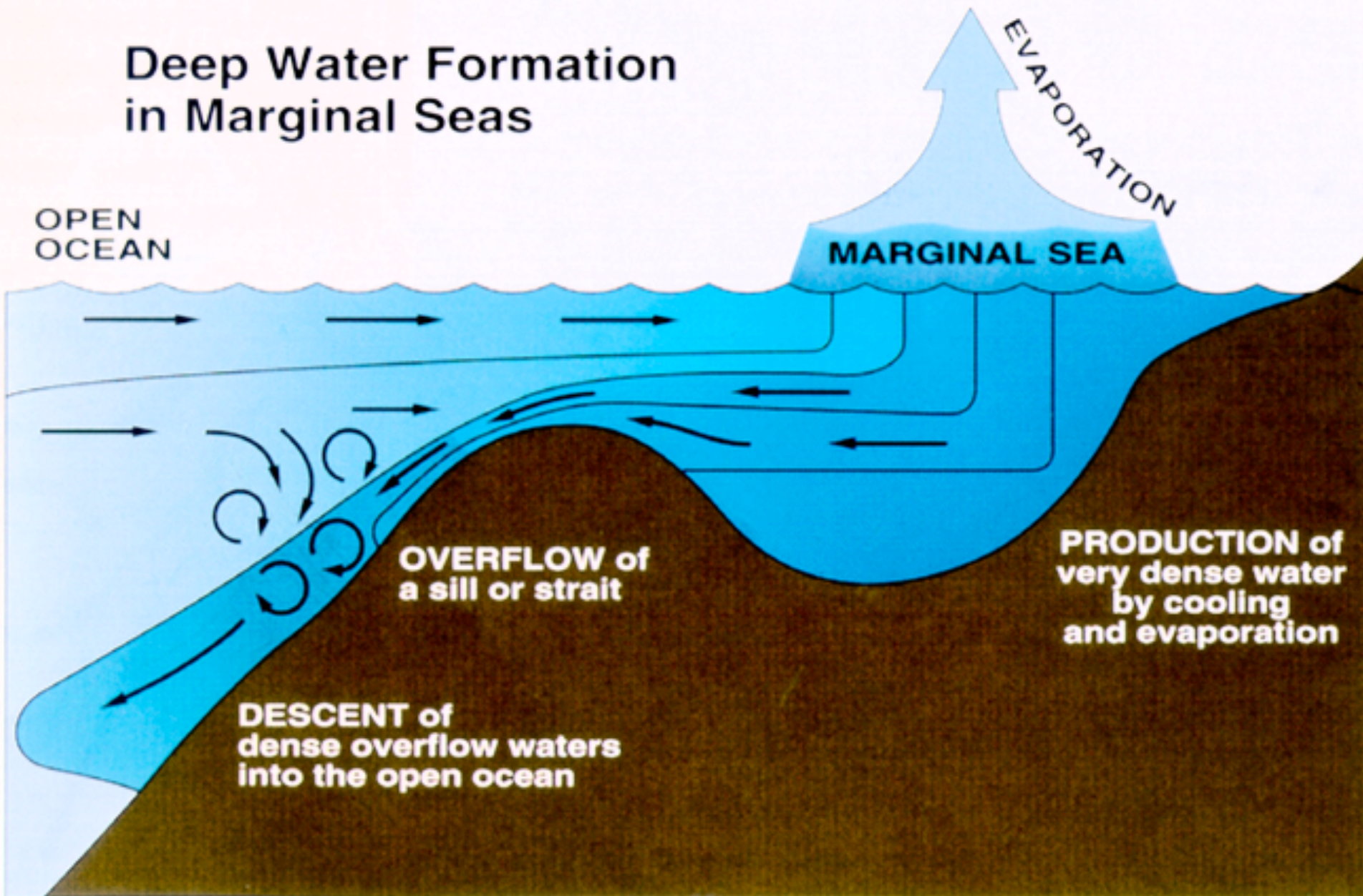
EVAPORATION

MARGINAL SEA

OVERFLOW of
a sill or strait

PRODUCTION of
very dense water
by cooling
and evaporation

DESCENT of
dense overflow waters
into the open ocean



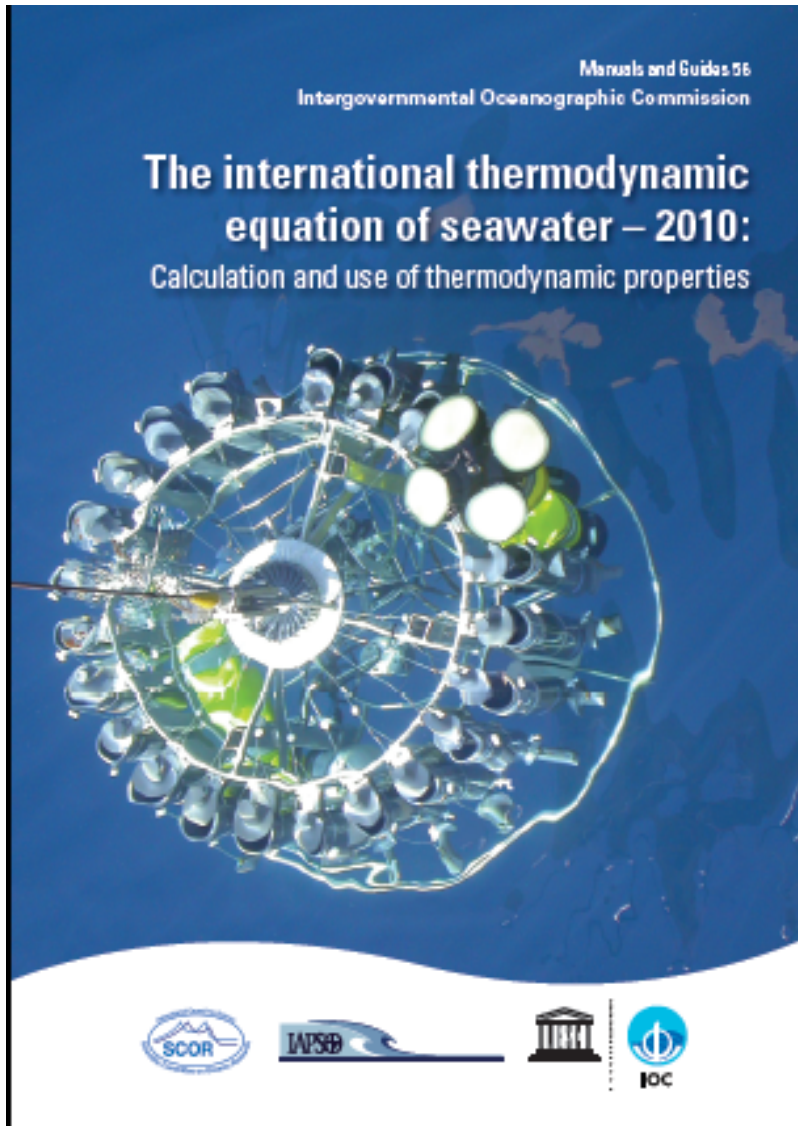
Outline

Horizontal (wind-driven?) circulation

- Some observations
- How does momentum from the wind enter the surface of the ocean?
- How do the surface layers affect the interior flow
- Western boundary currents

Vertical (density driven?) circulation

- Some observations
- **Digression: the equation of state**
- The overturning circulation
- Deep western boundary currents
- The importance of mixing



New 2010 UNESCO-endorsed standard for thermodynamic properties of seawater
www.teos-10.org replaces old UNESCO 1981 equation of state

Formulated in terms of a Gibbs function $g(S_A, T, p)$

S_A = Absolute Salinity (g/kg)

T = Temperature (K)

p = Pressure (Pa)

Advantage: Gives access to all known thermodynamic quantities

$$v = \frac{1}{\rho} = \frac{\partial g}{\partial p}$$

Specific Volume

$$\eta = -\frac{\partial g}{\partial T}$$

Specific Entropy

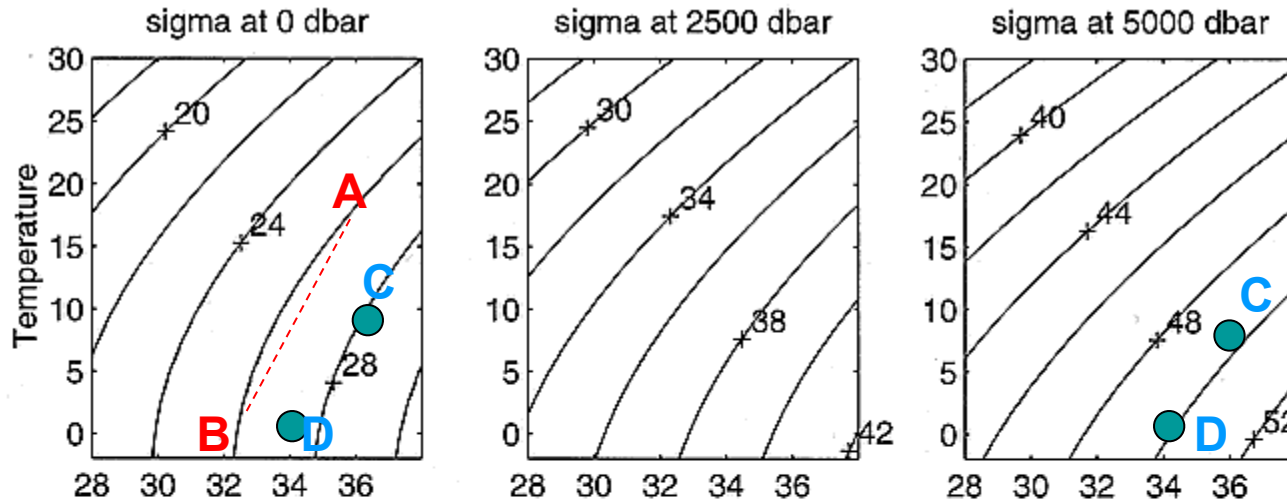
$$\mu = \frac{\partial g}{\partial S}$$

Chemical Potential

+ many more!!! (See TEOS-10 Manual)

Properties of equation of state

The density of seawater is a nonlinear function of temperature, salinity and pressure



Contours show potential density as function of temperature and salinity at different pressures (1 dbar = 1 m depth approx.)

Salinity variations affect density more at low temperature than at high temperature.

Cabelling: mix two water masses **A** and **B** of the same density, and the resulting water mass is denser.

Thermobaricity: two water masses **C** and **D**. At 0 dbar C may be denser than D. But move them adiabatically to 4000 dbar and D is denser than C. Need to allow for this in models or can get spurious static instability.

Outline

Horizontal (wind-driven?) circulation

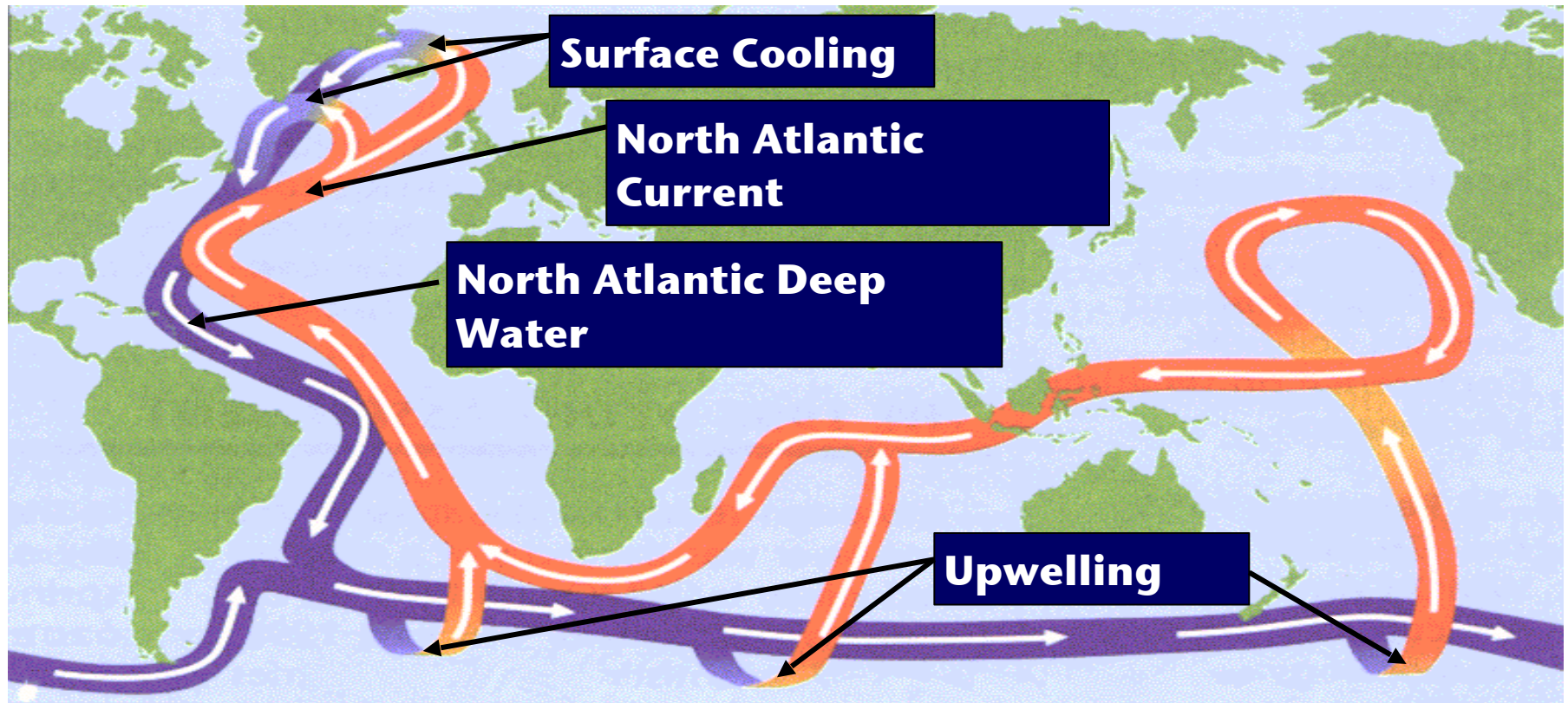
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The 'global conveyor belt'

or 'Thermohaline Circulation' THC or 'Meridional Overturning Circulation' MOC

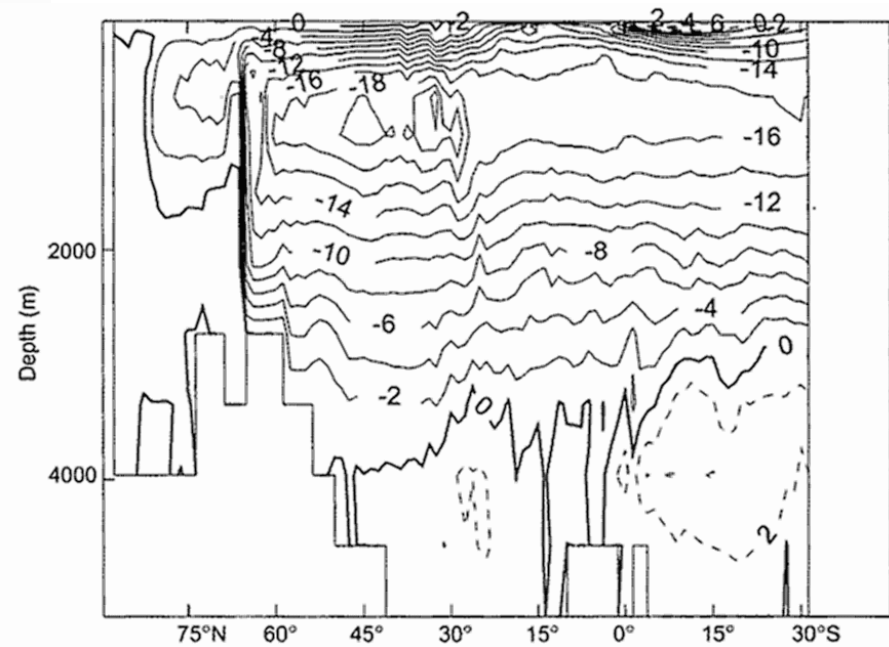


Surface heat exchange is a 'driver'

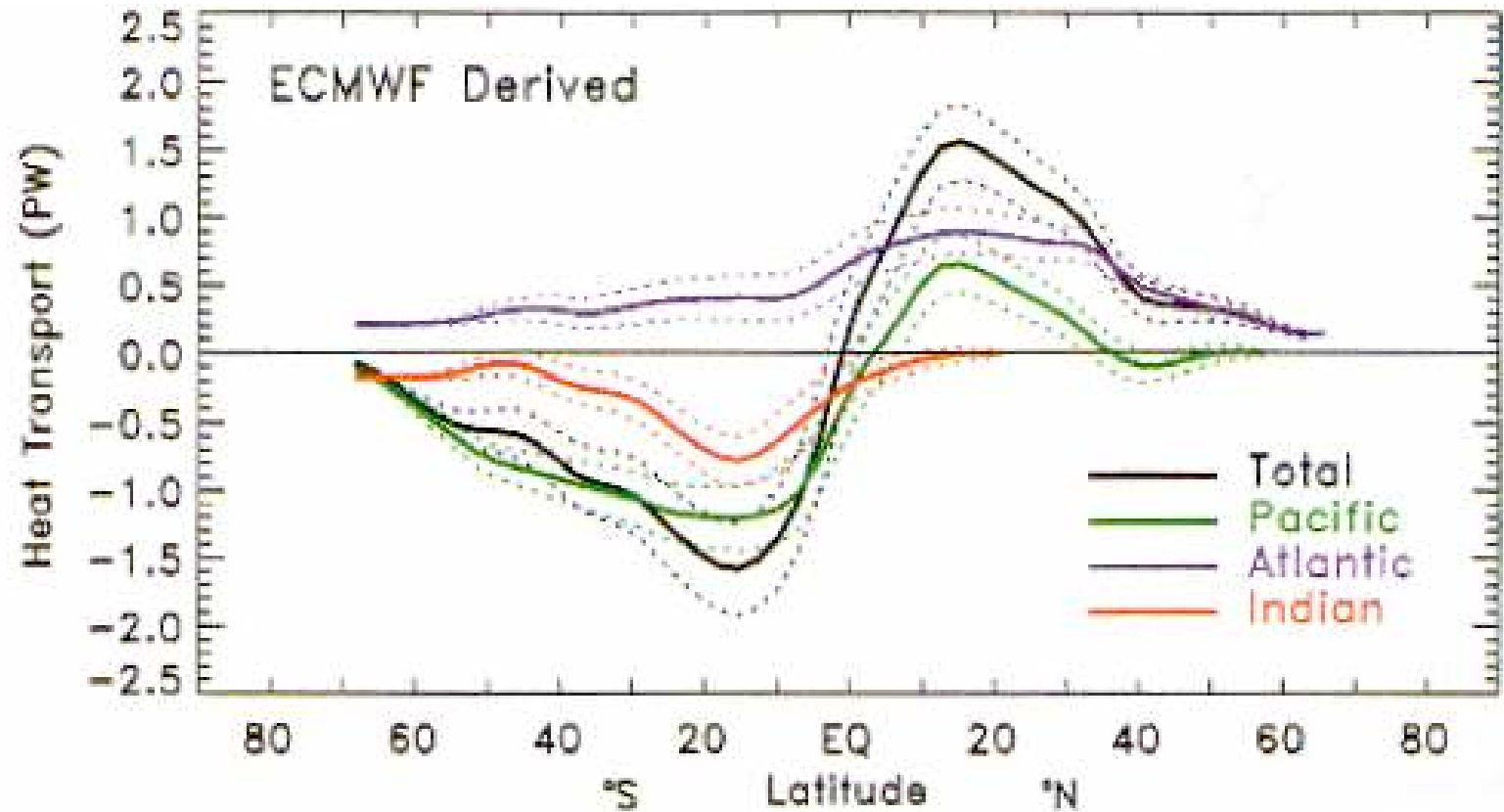
Role of salinity is more mysterious: transient brake, long-term accelerator?



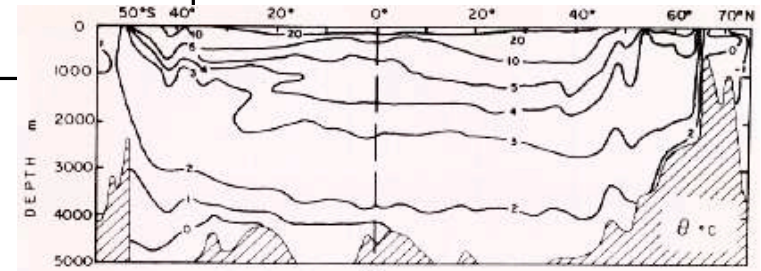
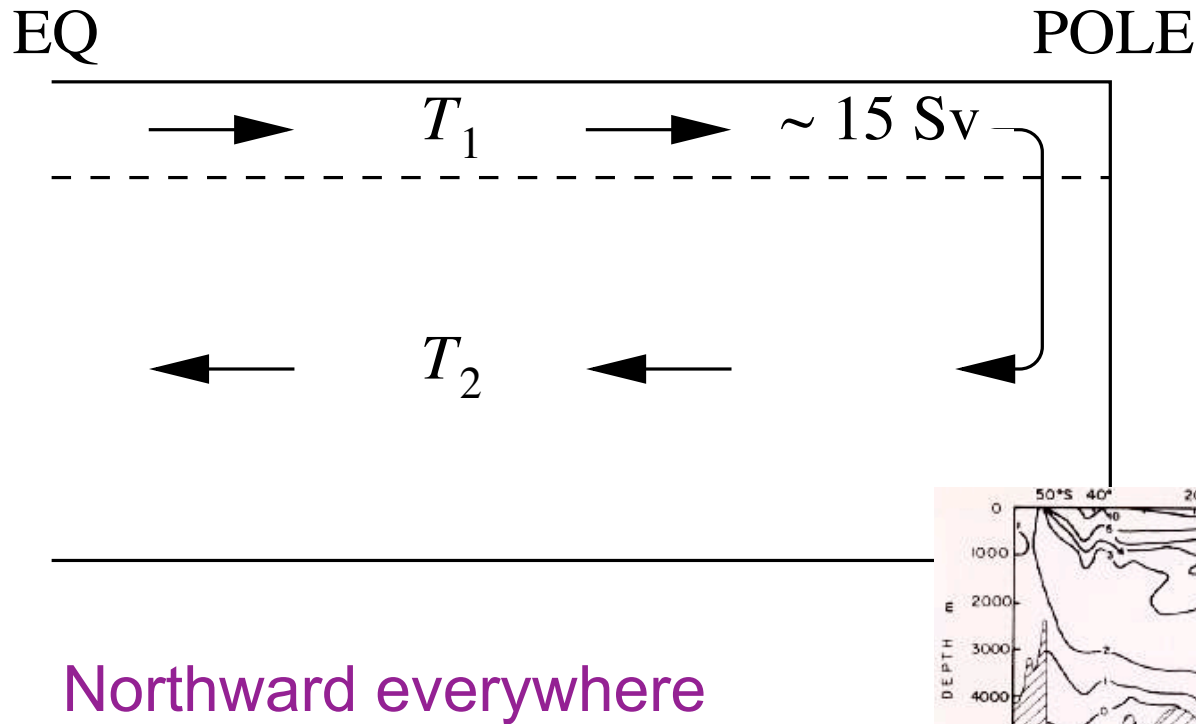
Ocean General Circulation
Model representation: zonally-
integrated meridional
overturning stream function



Poleward heat transport



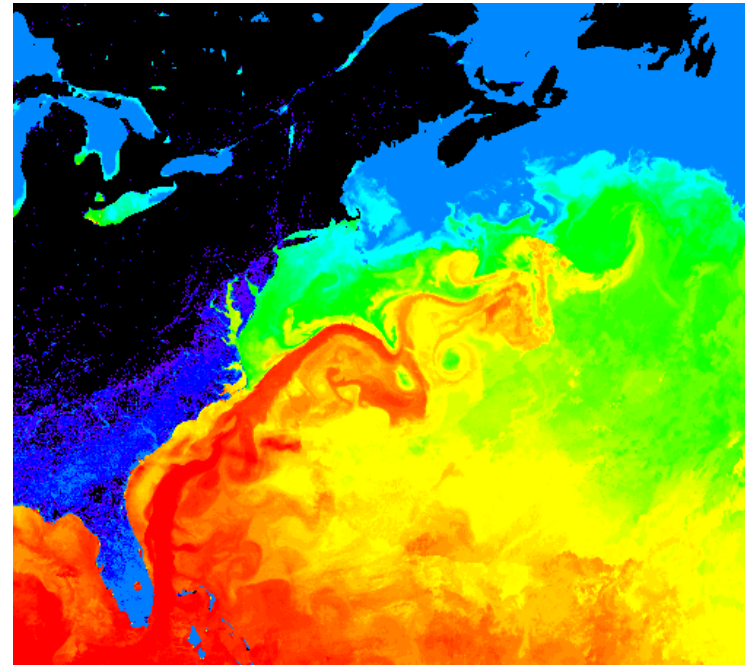
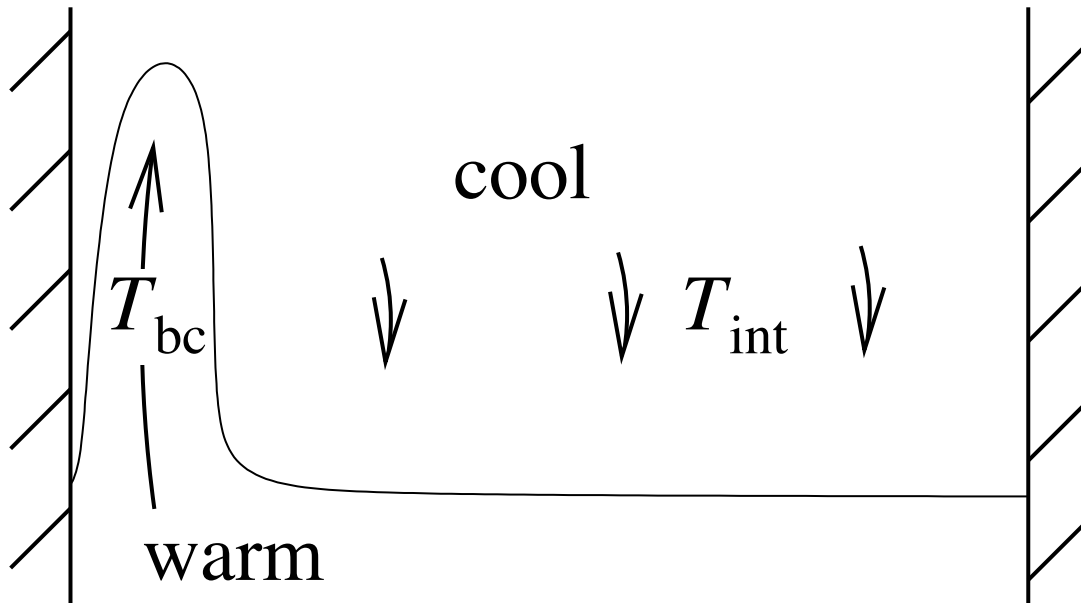
Poleward Heat Transport



$$\begin{aligned}
 (\text{heat transport}) &\sim (\text{volume transport}) \rho_0 C_w (T_1 - T_2) \\
 &\sim 15 \times 10^6 \text{ m}^3 \text{ s}^{-1} \cdot 10^3 \text{ kg m}^{-3} \cdot 4 \times 10^3 \text{ J K}^{-1} \text{ kg}^{-1} \cdot 15 \text{ K} \\
 &\sim 0.9 \times 10^{15} \text{ W} \\
 &= 0.9 \text{ PW}
 \end{aligned}$$

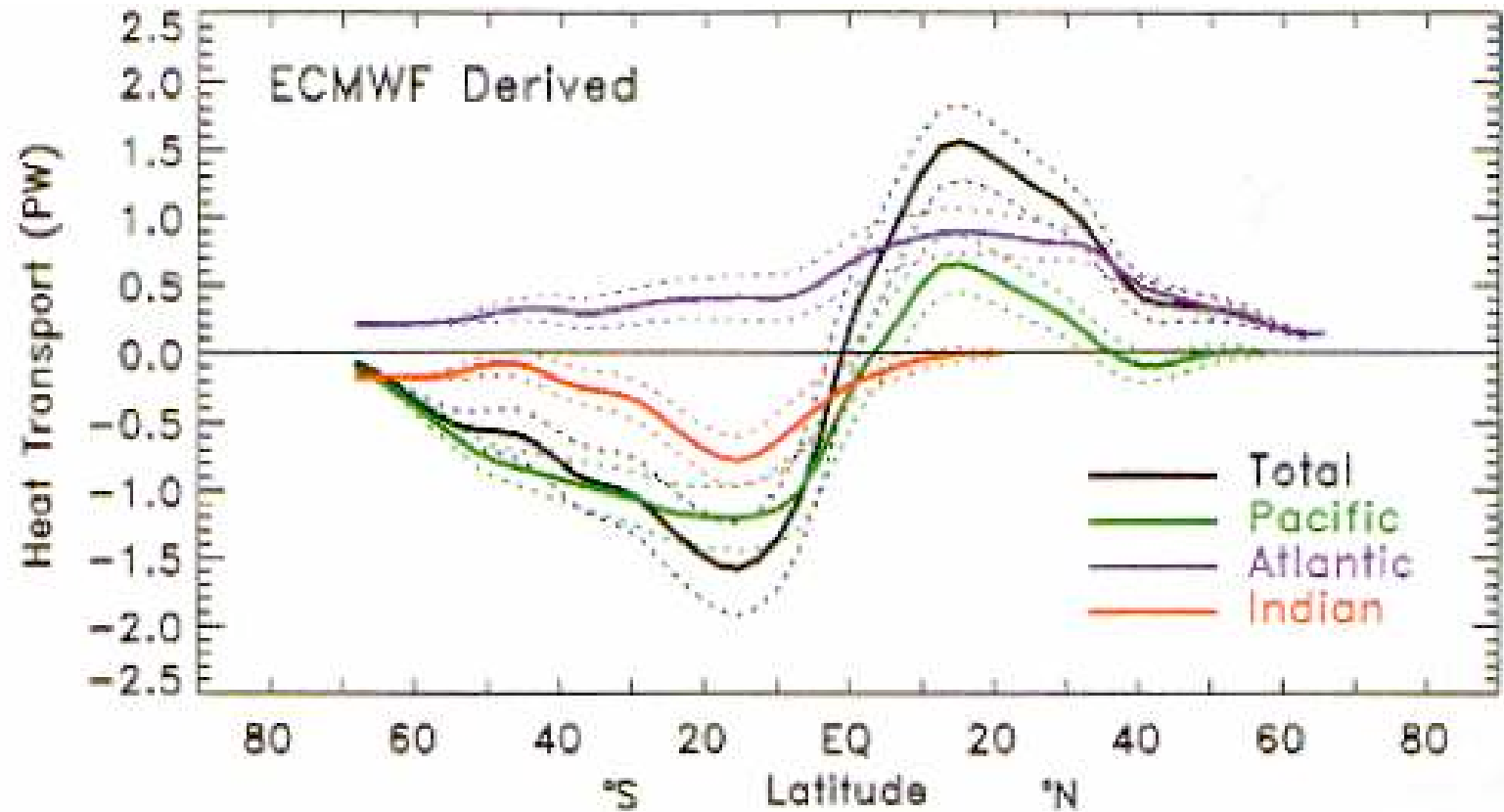
Poleward Heat Transport

$$\begin{aligned}(\text{heat transport}) &\sim (\text{volume transport}) \rho_0 C_w (T_{bc} - T_{int}) \\ &\sim 30 \times 10^6 m^3 s^{-1} \cdot 10^3 kg m^{-3} \cdot 4 \times 10^3 J K^{-1} kg^{-1} \cdot 5 K \\ &\sim 0.9 \times 10^{15} W \\ &= 0.6 PW\end{aligned}$$



Poleward in every hemisphere

Poleward heat transport



Water mass transformations in the Nordic Seas and Subpolar Atlantic



Note cold, deep flow escapes through narrow (20 km) gaps in Greenland-Scotland ridge (how to model??) and flows southward in a deep western boundary current

Outline

Horizontal (wind-driven?) circulation

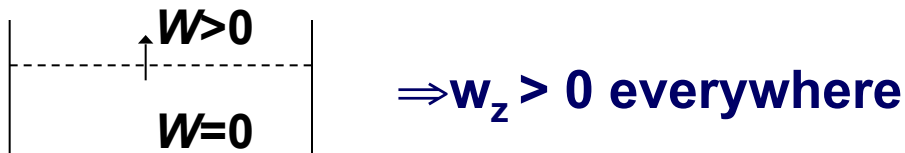
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Why is the southward flow in a deep western boundary current?

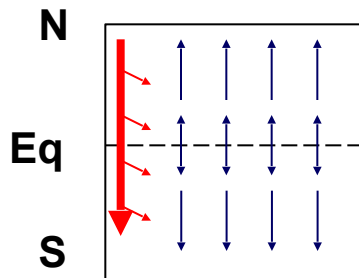
Assume the upwelling part of the circulation is uniform everywhere in the interior (NOTE: it isn't!), and bottom is flat. Consider flow in bottom layer:



In geostrophic interior get linear vorticity balance:

$$\beta v = f w_z$$

$\beta > 0$ everywhere, but f changes sign according to hemisphere. So v must be < 0 (S.Hemi) and > 0 (N. Hemi), i.e. *poleward and divergent* everywhere. So interior flow can't provide the mass needed to supply the upwelling. The mass must come from an (ageostrophic) deep western boundary current.



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Importance of vertical mixing in closing the MOC

Consider heat balance of a region in the interior thermocline.

Upwelling cools the region. This is balanced by downward mixing of heat:

$$w\theta_z = K\theta_{zz} \quad [\theta = \text{potential temperature, } K = \text{vertical diffusivity}]$$

Integrate to get

$$(*) \quad \theta = C + D \exp[(w/K)z]$$

Observations suggest that about 20 Sv of North Atlantic Deep Water are produced ($1 \text{ Sv} = 10^6 \text{ m}^3 \text{ s}^{-1}$). Assuming this upwells uniformly over the whole ocean gives w of order 10^{-7} ms^{-1} or **1 cm day⁻¹**.

Now fit eqn (*), with this w , to the observed mean temperature profile. Gives

$$K = 10^{-4} \text{ m}^2 \text{ s}^{-1} \text{ or } \mathbf{1 \text{ cm}^2 \text{ s}^{-1}}$$

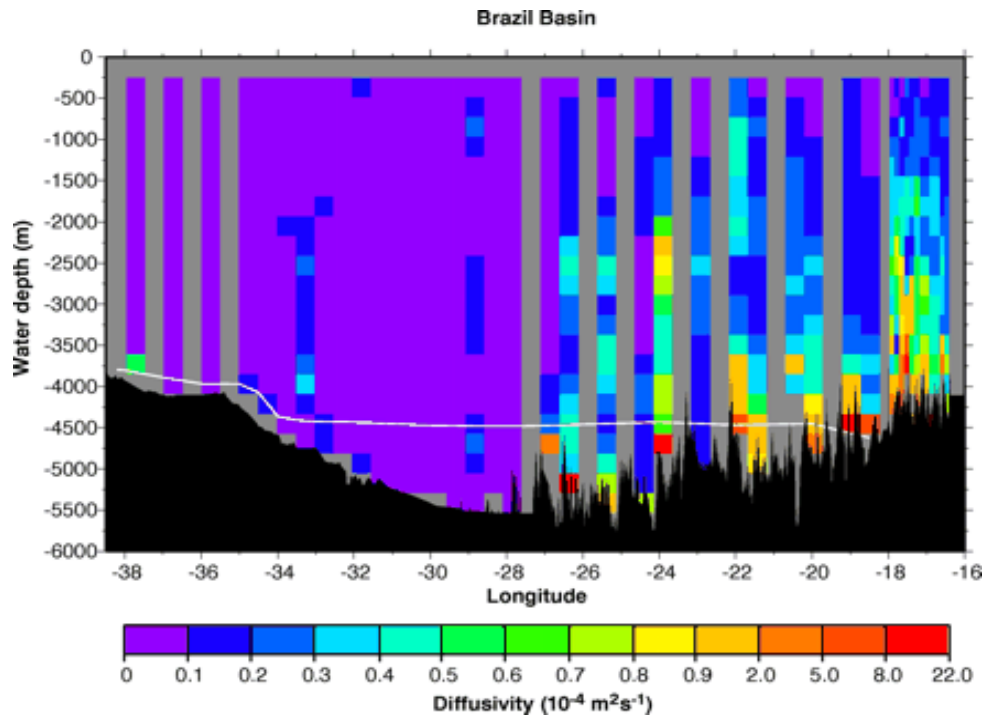
The vertical mixing is crucial to close the MOC – more mixing (K) allows stronger overturning (w) [seen in GCMs, e.g. F. Bryan JPO 1987].

Mixing in the real ocean

Munk's 'abyssal recipe' tells us how much mixing there must be in the ocean.

Tracer release experiments into the thermocline always produce a value of K around $0.1 \text{ cm}^2\text{s}^{-1}$, $\ll$ Munk's value. What's going on?

Vertical mixing increases the gravitational PE of the water column and so requires some mechanical energy input. Where does this come from? (current 'hot topic', see e.g. Munk and Wunsch)



Direct estimates of K in South Atlantic. Note hot spots near rough topography: internal wave breaking?

Polzin et al 1997

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Final Exercise: see how many of the features noted on the observational pictures in this lecture, you can explain.

Try to explain some other features of the observations!